

LASERTEC CORPORATION (TOKYO: 6920)

Japan's Most Actively Traded Stock Is A Colossal Fraud And Ticking Time Bomb, Reminiscent of Olympus, Toshiba, And Other Infamous Accounting Scandals

- One of the largest corporate frauds in Japanese history, a textbook accounting scheme faking revenue, margins, and earnings, with a looming impairment charge that will wipe out most of its earnings since the launch of its flagship ACTIS EUV inspection machine – the driver of its \$23B market cap and improbable rise from a penny stock to one of the best performing and most actively traded stocks on earth
- The ACTIS system is defective with a fatally flawed EUV light source and other insurmountable problems, which it has concealed from investors - the absurdity of an EUV concept stock which has no functioning EUV technology
- An alarming level of inventory that is the highest of any semicap tools supplier in the world, a sign of fraudulent accounting and the core of its accounting charade
- The “new” ACTIS A300, recently announced as its next-generation EUV product and driver of growth, is a hoax – a cover story to conceal flaws in its flagship A150 EUV tool and its desperate recall and replacement effort
- The company's new Innovation Park in Yokohama is a brazen fraud, a purported R&D and production campus with two supposed “fabs” that is 5X the size of its headquarters and the next phase of its expansion, announced in 2022 as the largest capital expenditure in its history
- Our investigators visited ~20 times from Dec 2023 to May 2024 and found a generally deserted facility with no R&D or production activity - contradicting the company's claims and suggesting its promotional pics of the “fab” are fake, based on extensive surveillance including site interviews, weeks of 24/7 time-lapse camera footage, and aerial drones
- Its largest customers – TSMC, Intel, and Samsung which are ~80% of revenue – are exasperated and furious to the point that TSMC has ceased buying its EUV machines, and the key Intel executive in charge of Lasertec's tools has been allegedly banned from meetings for “blowing up” on Lasertec due to his anger at the tool's failure
- KLA's imminent EUV market entry will be “catastrophic” for Lasertec, with customers “crying” and “begging” KLA to hurry – “devastating”; “game over”; “everything could turn around in a second...they're out of business, basically”

“To hide a lie, a thousand lies are needed”

嘘を隠すには、もっと大きな嘘が必要になる

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Detailed summary / key findings

1. Introduction to the real Lasertec Corporation

- Lasertec was a small, obscure semiconductor equipment supplier that struggled for a decade as a penny stock, trading at 1-2X revenue with a \$50-100MM market cap and no growth – and now suddenly sports a \$23B valuation, hailed as “Japan’s ASML.” and one of the most actively traded stocks on earth. It spent almost nothing on R&D yet claims to have a monopoly with its actinic EUV mask inspection tool, magically replicating a technology that took ASML 20 years, \$10B in investment, and 1,000 engineers to develop. We conducted an intensive 5-month investigation encompassing over 20 research interviews and ~20 investigator field visits to its new R&D campus in Yokohama, leading us to conclude that Lasertec is one of the largest frauds in Japanese corporate history.

2. Lasertec is an extreme, textbook accounting fraud that checks off every box: claiming to have the highest margins in the industry; but alarmingly little cash flow and the lowest cash conversion among its peers; accompanied by the highest inventory level of any semicap equipment supplier in the world. Overstated revenue and earnings are at the core of the fraud, offset by improperly valued inventory, with a looming, inevitable 65-70% write-down and impairment charge – a loss that will wipe out ~70% of its cumulative earnings since the launch of its flagship ACTIS EUV system and 85% of its retained earnings. The hit will far exceed what Olympus incurred and is considerable versus Toshiba, when their frauds were exposed.

- Lasertec is a classic Asia accounting fraud explained by a simple framework, whereby overstated revenue and fake earnings are offset with unusual, unexplained large increases in balance sheet accounts; accompanied by unusually high reported margins, but with no corresponding cash flow. A fraudster is hence trapped by double entry bookkeeping and the interdependence between the income statement, balance sheet, and cash flow statement. The typical warning sign that a forensic accountant therefore looks for: are there any unusual assets or unexplained large increases in assets such as inventory, particularly relative to revenues?
- The key point: overstated revenues and/or understated expenses will result in overstated retained earnings and must be offset by an overstatement of assets, understatement of liabilities, or understatement of another component of equity. For example, if a company sells goods (inventory) and reports the revenue on the income statement but fails to transfer the cost of that inventory (an asset on the balance sheet) to cost of goods sold (an expense on the income statement), earnings and assets in the form of inventory both become overstated. Similarly, if an asset - such as finished goods inventory - has declined in value, the company can continue to show it on the balance sheet at the older, higher cost or value. Thus, spiking inventory is associated with an overstatement of income by understating cost of goods sold – leading to margins that are unusually high.

Detailed summary / key findings

2. Lasertec is an extreme, textbook accounting fraud (cont'd)

- Effectively, the company is deferring reporting inventory expenses to a later period. The inventory account will increase rapidly over time and keep getting larger and larger. In frauds there is a limit and eventually the bubble must burst – one cannot outrun the fundamental accounting equation forever. Eventually everything must balance. The moment of truth typically involves an impairment charge and restatement that wipes out the previous years of revenue and/or profits, thus exposing the details of the fraud.
- Lasertec suddenly fired their longtime auditor Deloitte Touche Tohmatsu in 2018, after using them for at least the previous 15 years – a red flag as auditors are typically only fired after they start asking too many questions. The timing is suspicious as the firing occurred at the end of the same fiscal year in which inventory metrics began their troubling spike. Lasertec provided no explanation for the abrupt substitution. We note further that the disclosure in Lasertec's filings for its inventory valuation methodology is opaque and unusually terse.
- Lasertec bears similarities to Olympus which was exposed in 2011 as one of Japan's largest accounting frauds, illustrating how an asset account is used as an offset for fake earnings. The scheme collapsed when the company was forced to recognize losses via an impairment charge – the same end game for Lasertec when it admits that earnings and inventory are fraudulently inflated and takes a similar, massive impairment.
- In Lasertec's case, the offset is highly anomalous inventory of ¥162B/\$1.1B, hitting 120-130% of revenue in some recent quarters and currently at 70% of LTM revenue. The inventory makes no sense and has never been adequately explained. Various measures illustrate its startling level: 1) it has barely fluctuated in the last 6 quarters, a statistical oddity typical of a falsified number; 2) it is highly unusual versus Lasertec's historical level of inventory to revenue and assets, accelerating in 2019/2020 concurrent with the its EUV market entry; 3) the inventory level is the highest of any public semiconductor equipment supplier in the world, of which there ~150 globally greater than \$500MM/ ¥78B market cap; 4) it is triple or quadruple the level relative to the 5 largest semicap equipment players (ASML, Applied Materials, Lam, Tokyo Electron, and KLA), which all make similar complex, long lead-time equipment with the same income statement and balance sheet characteristics

Detailed summary / key findings

2. Lasertec is an extreme, textbook accounting fraud (cont'd)

- As an additional indication that the inventory account makes no sense, we detail the striking, recent breakdown in the historical correlation between inventory and “advances,” one of the largest line items in its balance sheet and its term for deferred revenue - a liability representing funds that have been purportedly collected but not yet earned as revenue. Even if a company isn't booking fictitious revenue, it may engage in aggressive revenue recognition and report a sale before it has occurred or before the revenue is earned. Deferred revenue is thus a high-risk category as a fraudster can manipulate deferrals to create current period revenue to meet expectations. Recent, unusual swings in Lasertec's deferred revenue are a major red flag, showing that, showing that basically all of its incremental revenue in the last 5 quarters – since Mar '23 – have come from siphoning the account, which is of course consistent with the lack of cash flow.
- Lasertec's inventory charade is not only brazen but hiding in plain sight – it claims that 100% of its \$1.1 billion in inventory is raw materials and work-in-process with 0% finished goods. Incredibly, its financial statements over the past 5 years record zero finished goods in any period – its balance sheet doesn't even include a line item for it. Misclassifying and improperly valuing inventory in an attempt to evade impairment charges that would wipe out historical earnings is a tactic common to some of the most significant fraud cases in history. Lasertec has no option but to claim no finished goods – its stock would have imploded if investors saw machines piling up.
- Lasertec's purported lack of finished goods inventory and reporting methodology is highly abnormal and lacks credibility versus the five largest semicap equipment players, for whom finished goods has averaged 30% as a percent of total over the last 5 years versus 0% for Lasertec – each one breaks out inventory by raw materials, work-in-process, and finished goods; several include additional disclosure for allowances and progress payments.
- It appears to be an open secret among Lasertec's investor relations team that inventory is being misclassified. We hired a local consultant to meet with IR in March 2024 at Lasertec headquarters, who at one point asked “And no finished goods in there, right?” The IR rep replied “No. All work in process.” while smiling with a knowing wink.

Detailed summary / key findings

2. Lasertec is an extreme, textbook accounting fraud (cont'd)

- Photographic evidence obtained by our investigators indicates that Lasertec is stuffing finishing goods inventory at its new “Innovation Park” and its so-called “fabs” - we show in a later section that this campus which is five times the size Lasertec headquarters is basically deserted with no production or R&D. During ~20 visits to the location, our investigators saw no activity except finished machines being delivered once or twice a week for storage, all hermetically sealed in protecting wrapping and clearly finished goods. Given that some of these machines sell for \$50MM each, Lasertec’s claim of 100% raw materials and work-in-process and 0% finished goods is clearly false.
- Lasertec’s dismal cash flow makes no sense relative to margins, which are simply too good to be true, as is typical in such frauds. Its EBIT margin has expanded dramatically: 10% (2010) to 28% (2019) when it launched its ACTIS EUV machine to 41% in 2023. It is an outlier on numerous counts: 1) Lasertec reports the highest EBIT margin of any semicap equipment company in the world with more than one billion USD of revenue; 2) the margin is significantly higher than every one of the five largest semiconductor equipment players – even eclipsing ASML’s, a \$350 billion market cap giant which invented EUV lithography; 3) its margins are so astonishing that they exceed Apple’s (30%), Google’s (29%), Meta/Facebook’s (37%), and are on par with Microsoft’s (42%).
- Lasertec’s gross margin and net income margin are just as vertigo-inducing as its operating margin. In each of last 5 years, its gross margin exceeded ASML’s and each of the 5 largest semicap companies except for KLA. Lasertec claims its net income margin has expanded significantly over the last five years, and in the last 12 months now exceeds even KLA’s.
- With margins far higher than the five largest semicap companies, one would expect Lasertec’s net income – the earnings reported on an accrual basis on its income statement – to convert to copious amounts of operating cash flow – and at a higher rate given its higher margins. But therein lies a key to the fraud – while Lasertec claims to have the highest profitability, its cash conversion is the lowest – a striking anomaly at the center of the largest corporate frauds in history. Cash conversion has plummeted since the launch of its ACTIS EUV line in 2019/2020 – over the last 5 years, one dollar of earnings has produced only 65 cents of cash, in contrast to its peers which have extraordinary cash conversion, such as ASML with 130-180% conversion despite margins far lower than the ones Lasertec claims; and KLA at >100% in all but one of the last 5 years

Detailed summary / key findings

2. Lasertec is an extreme, textbook accounting fraud (cont'd)

- Lasertec's cash conversion has dropped sharply over time, particularly over the last two years – collapsing even further in the last two quarters (30% Dec'23 and 28% Mar'24), in addition to exhibiting troubling volatility, a typical progression in frauds with a high-flying stock and pressure to show rising earnings, which requires the scheme to get larger every period until it becomes unsustainable. In the 5 year period preceding its ACTIS EUV line (2014-2018), cash conversion was 90% - versus only 16% and 52% in 2022 and 2023 using a rolling average. In 3 of the last 4 quarters, the conversion was 26-30% - indicating that 70% of reported net income is just accrual estimates and assumptions in a spreadsheet versus cash.
- It's extraordinary that Lasertec has a \$25B market cap yet only \$197MM of cash on its balance sheet – flat over the 5 years since the launch of its ACTIS EUV product, despite 40% margins. Over the same period, peers like ASML, Applied Materials, Lam, and KLA have seen their cash reserves surge – despite lower margins compared to Lasertec – even after repurchasing billions in stock and funding growth and inventory. Shockingly, Lasertec generates so little cash that it can't seem to fund its dividend without debt – putting it at risk.
- Other cash flow-based ratios are also useful in assessing quality of earnings – and confirm the problem at Lasertec. We compute an operating cash margin to show the percentage of revenue reflected in operating cash flow; as well as a cash return on assets. Lasertec spins to investors that it is an “asset light” or “fab-light” company: “most of our production is outsourced to third parties under [a] fab-lite strategy.” Its cash return on assets should therefore be far higher, but it is only about half that of ASML and KLA – and has also collapsed over time. Each dollar of revenue at Lasertec similarly produces far less cash than its peers.
- Manipulating inventory to falsify gross margins and earnings is an easy temptation for a semiconductor company trying to meet lofty market expectations. The potential for abuse is highlighted in the audit opinions at several of the largest semicap players. Auditors for both KLA and Lam recently noted inventory valuation as the one and only Critical Audit Matter in their last audit opinions, a designation reserved for the highest-risk and most material areas in the financial reports. Inventory has been at the core of large-scale fraud at various semis players, such as Toshiba's semiconductor unit and MagnaChip, a Korean company which trades on the New York Stock Exchange and which settled fraud charges with the Securities and Exchange Commission in 2017: “MagnaChip recognized revenue on sales of incomplete or unshipped products...and delayed booking obsolete or aged inventory to manipulate its gross margin.” The SEC detailed how Magnachip “overstated revenue for nearly two years in response to immense pressure...each quarter to meet revenue and gross margin targets that had been communicated to the public...a panoply of accounting tricks to artificially meet financial targets.”

Detailed summary / key findings

2. Lasertec is an extreme, textbook accounting fraud (cont'd)

- Lasertec's accounting scheme is rapidly approaching the final stage, which is usually preceded by unusual tremors – ignored or misunderstood by most – as pressure builds before the inevitable earthquake. A major tremor occurred in the quarter ended Sep 30, 2023, when Lasertec's gross margin mysteriously plunged from 56% to 36%, magically recovering to 56% the next quarter. The company's explanations were evasive. Unexpected, dramatic anomalies in the financial reports are a tell-tale sign that a company is struggling to hold a scheme together.
- The gross margin collapse, for reasons we detail, is due to the fact that Lasertec's EUV system is defective and unreliable to the point that Lasertec has no functioning EUV technology; and that key customers like TSMC and Intel are exasperated, angry, and refusing or delaying shipment and/or acceptance of Lasertec's tools. Lasertec never explained what was different about the EUV tools sold that quarter such that they had no margin – the obvious implication, however, is that if those systems are worthless, then so is the \$1.2B of inventory on its balance sheet, which is therefore mis-valued to avoid an impairment charge.
- Color from key Lasertec customers corroborates that the EUV inventory is worthless and therefore mis-valued, with the company quietly offering discounts of up to ~75% on its flagship ACTIS EUV tools, as well as other inducements and concessions. The customers indicated that Lasertec is playing inventory valuation games: “I can tell you right off the bat...they haven't taken any impairments against their balance sheet, so they're probably over-valuing their inventory far greater than it's actually worth...”
- The amount by which Lasertec is fraudulently valuing inventory – and therefore earnings, as well as the magnitude of the impairment charge it is trying to avoid – is straightforward to deduce. We use 5 simple methods, all of which arrive at a similar figure: an inevitable 65-70% write-down of its \$1.1B / ¥162B of inventory as of Mar '24 – a ~\$700-750MM / ¥105-113B loss that would wipe out ~70% of its cumulative earnings since the launch of its ACTIS product in late 2019 through the most recent quarter. The loss and retained earnings hit we think is imminent far exceeds what Olympus incurred in 2012 in the immediate aftermath of its scandal (¥49B/\$300MM and ¥53B/\$460MM, respectively); and is considerable relative to Toshiba's loss and retained earnings hit of ¥400B/\$2.6B and ¥378B/\$2.4B in 2014/2016 – a sobering fact given Lasertec is only 3.5% of Toshiba's revenue of ¥6,300B/\$40B at the time

Detailed summary / key findings

3. The origin of the accounting fraud: Lasertec's flagship ACTIS EUV product line is defective with a fatally flawed EUV light source – the core of any EUV system – in addition to insurmountable problems with other key components like the optics and mask stage. Shockingly, it has a \$25B market cap as an EUV story but lacks an EUV technology that actually works. Its inspection tool is an unmitigated disaster due to a) contamination and debris from tin splatter; b) instability and flickering which leads to false positives and other mask inspection failures; c) onerous cleaning and maintenance that leads to unacceptably low reliability, uptime, and throughput; and d) a short lifespan that drives frequent replacement and a burdensome cost of ownership. The failures have wiped out its margin – 3-4 full-time field service engineers per each machine for repairs and troubleshooting; frequent, costly replacement of optics components; a covered-up recall effort to swap out the source; discounts and other concessions to angry customers - thus forcing it to fabricate its earnings and the value of \$1.1B of worthless inventory.

- The ACTIS EUV system is plagued by fatal levels of contamination, which results from the ignition of molten tin into plasma prior to the production and collection of EUV light. An ex-KLA executive in a leadership role in their EUV mask inspection group – who has extensively interacted with Lasertec and is intimately familiar with their technology - described “molten tin that splashes everywhere” with “puddles in the air” that “coat everything,” with “no way out” of the problem via attempts at “debris mitigation.”
- Contamination from tin-produced plasma is a well-known and long-standing obstacle in the scientific literature on EUV light source development, as igniting tin is the only way to produce the correct wavelength at the right power. The problem mainly afflicts the laser-assisted discharge-produced-plasma (LDP) approach employed by Lasertec, in contrast to ASML's laser-produced plasma (LPP) where the plasma is better controlled. Various papers and studies detail the tin and particle contamination problem that plagues Lasertec's EUV light source: the tools “must find defects but not contaminate!” and must be “the cleanest machines among all tools used in a fab/mask shop”; “plasma based sources are the dirtiest parts in EUV tools...”
- The seriousness of the contamination problem was confirmed by key Lasertec customers such as TSMC and Intel; multiple Lasertec field service engineers; Lasertec's EUV source technology suppliers such as Ushio and Isteq; and numerous other experts. They described Lasertec's source as “ridiculous” and inherently flawed and stated that they “made an awful mistake” as they were desperate to launch a product given pressure from Intel: “they took a source without understanding the complexity...and the problems associated with damage from debris...they did this because it was the only available source.”

Detailed summary / key findings

3. Lasertec's flagship ACTIS EUV product line is defective with a fatally flawed EUV light source (cont'd)

- Lasertec's EUV source "is terrible" and "creates a mess" given its use of discharge electrodes - similar to car spark plugs – and propellers splashing in molten tin, in contrast to ASML's superior laser-based solution as well as the xenon-based source KLA is developing.
- Off-gassing from tin plasma and volatile organic compounds are additional sources of mask contamination, undermining the machine's sole purpose of mask inspection: "one is particles...two is VOC...off-gassing...which coats the optics and produces damage to the optical surfaces...the particles propagate and eventually find themselves on the mask and you find these specks." Particles generated by Lasertec's machine leads to "degradation of the imaging optics...and particulation on the actual mask," which "you cannot clean" as it coats the source of the EUV light": "gradually you lose it and you have to replace it."
- Lasertec's particle contamination problem makes it highly problematic for P2 mask inspection – the only step in the inspection process where its EUV tool can be used. Mask inspection involves four steps, each with distinct inspection tools: P0, P1, P2, and P3. as we shall explain. However, since Lasertec's EUV mask inspection tool introduces new particle contamination in P2, key customers like Intel and TSMC are forced to then do a redundant inspection with the same KLA Teron tool used in P1 – defeating Lasertec's entire purpose and creating issues prior to the final step, called P3 which refers to inspection after a thin protective membrane called a "pellicle" is placed on the mask. As a result, Lasertec "customers are furious because they need to maintain another KLA tool as a backup on P2."
- After particle contamination, the second fatal problem with Lasertec's EUV light source is instability and flickering, which leads to false positives and other mask inspection failures. The light source is inherently unstable for various reasons, such as electrode erosion that plagues their discharge-produced plasma approach. We interviewed four on-site field service engineers tasked with addressing problems with Lasertec's actinic mask inspection (ACTIS) machines and the EUV light source in each one. Three of the engineers were recently employed by Lasertec and one is currently at Ushio to specifically address EUV source failures. All four confirmed that the machines were so instable that each one required 3 full-time Lasertec field engineers, for a total of 9 just in Intel's mask shop alone, which they stated has 3 Lasertec actinic inspection tools.

Detailed summary / key findings

3. Lasertec's flagship ACTIS EUV product line is defective with a fatally flawed EUV light source (cont'd)

- The field service engineers provided color that was consistent and across all key customers such as Intel and TSMC: "across the three machines, we'd have daily issues" and that they were forced to "sacrifice" throughput and functionality "for maintaining uptime." Intel's fabs in Oregon experience the same problems with Lasertec's machine as the mask shop in California. The engineers corroborated that Lasertec's EUV source is unstable due to electrode erosion: "a problem because the tin when it splashes between those two fans and the discharge has electrodes" which "have erosion and therefore it's not stable and it's changing all the time." An ex-field engineer stated he "was hands-on the ACTIS line" and that "it was basically like a flickering candle": "having high repeatability became a big challenge"; "the light drifted out of focus or too far out that Intel engineers couldn't step in and resolve it...."
- Another former Lasertec application engineer, who was primarily allocated to Intel, stated that "the ACTIS A150...has a lot of problems" especially "regarding the chamber structure and the source" which "tends to be unstable": "customers want better maintenance and stability...the problem was the light source...it requires more time for maintenance once the machine breaks down."
- The Ushio engineer stated that "Lasertec causes a lot of headaches for their end user" and that each machine requires "a lot of work" in order "to keep it running": "on the light source side we have some issues...things break." He stated that "the struggle is to reduce" the light source downtime, and parts that need to be replaced;" that the source fails rapidly and has to be replaced every 2 to 4 weeks; that the light source is compromised to run "dimmer" to extend its lifetime by a week or two; that it breaks down faster in the mask shop because it has to run "at a much higher brightness"; and that Lasertec is "lucky that the competitors didn't jump on making pattern inspection machines right off the bat."
- Ushio detailed the mask inspection problems and failures caused by the EUV light source's instability and lack of brightness and power: "they get a dark frame or they get a false defect...it's light source fluctuation...the brightness is just going to fluctuate just as a function of the physics of the plasma." He stated that the tin-based plasma reaction is poorly controlled: "the plasma has a lot of freedom to move...that causes fluctuation...a difficult thing to control"; "it's hard to get enough power" to "get the brightness up...as a result the plasma is smaller...that can create problems with not getting enough illumination into the tool." With insufficient light source stability and brightness, "you'll get a frame where there's just not enough light and uniformity...you're going to have dark area and that's going to cause a false defect to show up."

Detailed summary / key findings

3. Lasertec's flagship ACTIS EUV product line is defective with a fatally flawed EUV light source (cont'd)

- A leading EUV lithography expert close to all major players including Lasertec, ASML, and KLA - as well as the industry's three key customers TSMC, Intel, and Samsung – described Lasertec's source as an unmitigated disaster. He stated that Ushio was secretive about the defects, but that he received some data and modeled it, which revealed it was fatally flawed: “I asked Ushio what was wrong with their source, and they wouldn't say...if your source is not stable, then the intensity of light coming in through it fluctuates...so you have a stability issue...they gave us the data, and we had about six to seven different models of Ushio data...the bottom line is that there is something wrong.” He was highly critical of Intel's credibility with respect to Lasertec, noting their large investment and need to save face as its sponsor: ““Intel came in and did their song and dance a couple of years ago, saying we've got Lasertec working...we're the winner...but three years later we're hearing the Ushio source doesn't work.””
- The third fatal problem with Lasertec's EUV-based inspection tool, in addition to contamination/debris and light source instability/flickering, is pitifully low reliability that leads to low uptime and throughput, compounded by onerous cleaning and maintenance. Multiple ex-Lasertec engineers indicated that the tools only work a little more than half the time, with 60-65% uptime, angering both Intel and TSMC. When we remarked that the utilization is poor, one engineer replied “Yeah. That kind of came with the territory.”
- We contrast Lasertec's low ~60% uptime with 96% for ASML's EUV source. In addition, ASML's EUV collector lasts for over a year, versus only a couple of weeks for the same critical component in Lasertec's machine. A KLA presentation on EUV mask inspection highlighted the absolute requirement for light source uptime: “Every % of tool availability is critical for mask shops that must run 24/7...anything less than 85% is not acceptable”; “the source has to be highly reliable to avoid impact to overall system availability.” They emphasized that plasma EUV is challenging as tin matter “must be ablated >5000 times per second...life-time is the biggest challenge”; “replacement of components (e.g. electrodes) every week is not an option...maintenance time of 5-10% represents a serious problem for the entire tool availability.”
- Several long-time industry executives close to and in contact with Lasertec's CEO provided information identical to that shared with us by field service engineers. One stated that his color on reliability issues comes directly from “close connections at Lasertec” but was apprehensive about sharing more: “...but I really can't go into more than that”; “some of the stuff I really can't share.” Lasertec's secrecy is a recurring theme echoed by its own engineers, which we shall return to. He continued that “Lasertec system reliability is pretty low...60% reliable...so it's not reliable enough...making chips work requires every step to be 99.9% reliable...the mask is super critical because if one mask is bad, every chip you make after that bad mask is bad.”

Detailed summary / key findings

3. Lasertec's flagship ACTIS EUV product line is defective with a fatally flawed EUV light source (cont'd)

- The reliability is so low that Lasertec's tool is simply not "production-worthy": "their tool doesn't work reliably"; "doesn't work well enough for production"; doesn't work well enough overall," which leads customers to beg KLA to hurry its competing actinic inspection tool – "and that's why customers want KLA to make a reliable tool."
- Lasertec's EUV light source is so plagued by the fatal problems just described - 1) contamination and debris; 2) instability and flickering; 3) low reliability, uptime, and throughput along with onerous cleaning and maintenance - that it has a short lifespan of only a couple of weeks before needing replacement and/or being sent to Germany for refurbishment, driving a burdensome cost of ownership. A former field service engineer stated "they were replacing them every, give or take, two weeks...being able to work on the system became rather challenging." He highlighted additional downtime from de-pressurizing and pressurizing the chamber, as EUV light is absorbed by air and glass and has to be enclosed in a vacuum: "...pumping up and pumping down was a significant amount of time...every time the chamber was opened or the light source was taken off, we had to de-vac...then pump it back up to vacuum."
- An industry executive in touch with Lasertec detailed the vast amount of cleaning and maintenance required to service and maintain the other optics besides the light source, such as the collector and mirrors that direct EUV light. He described seeing piles of Lasertec EUV collectors in Germany awaiting cleaning at a specialized vendor: "you replace the collector and send it to Germany"; "I've seen a bunch of these unbelievable amounts of collectors there from Lasertec"; "it costs hundreds of thousands of dollars" to clean each collector every month; "takes almost two days to chisel all this debris." The expert suggested the annual cost to clean each collector is \$2.5 million: "I'm telling you, the problem...started two years ago and they said that we cannot ship...they are not shipping because it's a liability...it's consuming all the margin."
- Our research interviews indicate that the net result of the contamination and frequent maintenance and refurbishing is that the entire optical system inside Lasertec's EUV inspection tool needs to be replaced at least every year: "...it coats the source of the EUV light until it develops a few mono-layers of coating...and gradually you lose it...and have to replace it": "I heard every year that they have to replace the optics...the refurbishing of those optics costs at least a million dollars or more...they need to replace mirror number two, mirror number four, and that could cost at least half a million dollars each."

Detailed summary / key findings

3. Lasertec's flagship ACTIS EUV product line is defective with a fatally flawed EUV light source (cont'd)

- Another longtime semiconductor industry executive based in Japan – who stated he is friendly with Lasertec's CEO – corroborated that their EUV source is defective with a short lifespan and low uptime. As a result, customers have to waste money buying duplicate back-up machines: “they had big trouble with the light source”; “they were struggling” because of the short lifespan “and then it needed replacement and after replacement they needed one month to tune the light source...so the tool was only available for maybe 60% or 70%...customers needed to buy more equipment than they originally planned to cover their requirements.” He elaborated that the EUV light source is “very low efficiency...in order to provide sufficient capacity, customers need to purchase more than one tool where they only would need one.”
- In addition to a defective light source and optics, the key mechanical component of Lasertec's actinic tool - the mask stage - is also fatally flawed. The stage is a critical component of both ASML EUV scanners that write the wafers as well as Lasertec's mask inspection machine, as it continuously moves and adjusts the wafer and mask into appropriate position in the x, y, and z planes during the writing or inspection process, requiring microscopic precision as EUV wavelength is only 13.5nm, a 10X higher resolution than previous DUV systems.
- The stage is a critical technology and key factor in ASML's dominance, as it developed a proprietary magnetic levitation system, while Lasertec was forced to buy an off-the-shelf, defective stage from Sumitomo – the only one available. The Sumitomo stage “is extremely complex and has a huge problem because of the contamination it produces”; “Lasertec chose the easiest route because it was available...you cannot buy a magnetic levitation stage”; “Lasertec cannot do it...that's the second thing they compromised, in addition to the light source.”
- The Sumitomo stage uses a crude pneumatic system that operates inside a vacuum, using pressure to move the mask versus ASML's magnetic levitation. The pressure forces and releases particles, causing contamination that ruins the mask, in addition to being complex with each stage costing millions: “it's very complex and very, very expensive...multiple millions...it uses pressure and then they try to suck it back up...it releases particles into the cavity of the optics...that's the number one source of particle generation .”
- The stage creates a third source of contamination - compounding the problems from tin splatter from the plasma reaction used to emit EUV light, and off-gassing: “the stage has a problem with particulation...all of them contribute to the degradation of the imaging optics and also the particulation on the actual mask”; “the stage has two problems: significantly high cost and complexity and...the fact that it produces particulation into the entire area” of the cavity of the tool, which consists of a large, open vacuum chamber that can't be partitioned with barriers as EUV cannot travel through air or glass.

Detailed summary / key findings

4. The “new” ACTIS A300, announced with fanfare in Nov 2023 as Lasertec’s next-generation EUV product and driver of growth, is a hoax – a cover story to conceal fatal flaws with the Ushio-supplied EUV light source in its flagship tool, the A150, and the resulting, costly recall and replacement effort as each source is ~\$10MM. The company’s EUV story - and its stock - would have collapsed had it revealed that the source – the entire basis of an EUV system - was defective and had to be hastily, secretly swapped with one from a small, 12-person Russian R&D project called Isteq, which it misrepresents as in-house Japanese technology under the name “Urashima.” However, the Isteq/Urashima source is just as defective as the Ushio one. The A300 is simply an attempt to mislead investors and spin a damage-control and recall effort as a “new” product. In addition, the much-hyped claim that the A300 is High NA-compatible – a term for ASML’s next-generation EUV lithography tool – is false as it has no such capability. Four different TSMC sources told us that the A300 recently failed their inspection process and that they have no intention of buying any, similar to color from other customers like Global Foundries and Marvell.

- The ACTIS A300 was announced on Nov 24, 2023, as the latest version of Lasertec’s actinic EUV mask inspection machine, replacing the A150, announced in 2019, as its flagship product. The key “innovation” in the A300 is a “new” EUV light source which Lasertec calls “Urashima.” The second upgrade is that the A300 purports to be High NA-compatible, which refers to next-generation EUV lithography with high numerical aperture (“NA”) for printing wafers at higher resolution, density, and smaller size. The Lasertec bull case hinges on growth from the new A300. Sell-side broker reports talk up the A300 as key to Lasertec’s next phase and its purported “monopoly” in EUV mask inspection.
- Rather than admitting that the Urashima light source is simply an old, re-packaged Russian EUV source no one else wanted, Lasertec is misleading investors by claiming it is a new Japanese-developed technology. Far from being a “new” and improved source, it is simply an old Isteq (LPP) laser-produced plasma technology developed over a decade ago. Lasertec could easily have licensed the Isteq technology prior to the ACTIS A150 launch in 2019 but chose Ushio instead. Given that there are only two or three EUV source suppliers to choose from, it is obvious that Lasertec would have evaluated Isteq over five years ago prior to rejecting it for Ushio – and are now desperately trying to cover up the Ushio failure and trying Isteq as a Hail Mary. We note various Isteq papers going back to 2013 on the Isteq LPP system.
- Isteq and its Russian parent RnD-ISAN jointly filed an international patent application for their LPP technology back in 2019/2020, and branded it as their “TEUS” EUV source product in a sales presentation in 2021, if not earlier – once again illustrating that “Urashima” – TEUS in disguise – is far from new.

Detailed summary / key findings

4. The “new” ACTIS A300, announced with fanfare in Nov 2023 as Lasertec’s next-generation EUV product and driver of growth, is a hoax (cont’d)

- A leading EUV lithography expert close to Lasertec, ASML, and KLA - as well as the three key customers TSMC, Intel, and Samsung – mocked the Urashima source and agreed emphatically when we asked if Lasertec was simply trying to cover up a light source failure and spin it as new, advanced product by using the old Isteq one: “That’s right. That’s right. That’s right. You got it right. You got it...this is supplier talk from Lasertec and also from Intel because they are spending a huge amount of money, too, and can’t look bad.”
- Multiple other research sources revealed to us that far from being a “new” product announced in November 2023, Lasertec began quietly replacing the Ushio source with the Isteq one approximately 18 to 24 months ago, sometime in 2022. An Ushio field service engineer based at Intel’s fab in Oregon stated that Intel has only about ten Lasertec machines in total in production – and that only three of these still have the Ushio source. The rest have the Isteq source now misleadingly labeled as “Urashima: “...the last one was a different light source; “they are not Ushio sources...I know they have a different light source...I know they announced the Urashima light source...I’ve heard Isteq thrown around as the creator of the IP for that source.” We note that the engineer appeared nervous at times while answering our questions, suggesting to us a coordinated attempt between Lasertec and its key customers to conceal that the “new” A300/Urashima announcement is essentially fake.
- A second research source - a longtime semiconductor industry advisor who consulted recently for Lasertec’s management - confirmed that the company already moved to the “new” Urashima source in 2022. Similar to the Ushio engineer, the expert was occasionally evasive in answering our questions and appeared uncomfortable when we asked bluntly if this means that the recent A300 product announcement is fake. He replied “That was a bit over a year ago” when we asked when Lasertec replaced the source – the interview was conducted in January 2024, and avoided commenting when we stated that the A300 is therefore not a new product: “Again, they are on a second-generation light source.”
- Multiple sources told us that the Urashima light source is as defective as the Ushio one and fails to address its fatal flaws. The Ushio field engineer based inside the Intel fab works closely with Lasertec to service the machines, giving him day-to-day visibility into all aspects of the technology. He stated that while “Lasertec tries to keep us in the dark” about the performance of the machines with the Urashima source versus the Ushio one, that the Urashima tools “are right next to our two”: “we can look...they can’t really stop us from looking.” He stated that the Lasertec engineers he interacts with work on both EUV light sources, and that “there’s as much downtime on those machines as there is on ours...”

Detailed summary / key findings

4. The “new” ACTIS A300, announced with fanfare in Nov 2023 as Lasertec’s next-generation EUV product and driver of growth, is a hoax (cont’d)

- We asked the Ushio engineer whether the A150 and Urashima A300 machines look any different when they are open, except for the two different EUV sources: “it’s all the same, as far as I can tell,” replying affirmatively when we then asked if everything else inside the tools is identical. He stated that Lasertec’s field engineers “complain” about the Urashima source as much as the A150 and that they are skeptical of their own company’s technology: “I work a lot with the boots on the ground at Lasertec, and they complain about having to maintain their light sources...[they] feel that management made some compromises....”
- The engineer indicated that the Urashima source is “down at least as often as the Ushio light source”; “they have the exact same inspection issues, so the Lasertec Urashima source has all the same problems”; “the light source has its own set of weaknesses” in addition to those that afflict the Ushio source; “I think that probably contributes to the downtime.”
- Although the Urashima source attempts to use laser-produced plasma (LPP) similar to ASML’s technology, where the plasma is better controlled, versus the Ushio source’s inferior laser-assisted discharge-produced-plasma (LDP) approach, Urashima is still plagued by the same inspection failures as Ushio. The plasma which emits EUV is just as unstable, leading to fluctuations and lack of brightness and power. The Ushio engineer provided details we find damning: “...the same results happen on their inspections” with Urashima; “they get a dark frame or they get a false defect...it’s light source fluctuation”; “in these light sources, the brightness is just going to fluctuate”; “difficult thing to control”; “hard to get enough power...to get the brightness up.”
- Experts we interviewed described the Ushio source as an unmitigated disaster and ridiculed claims that the Isteq source is superior as coming from the “same people” – presumably Lasertec and Intel – who made similarly false claims about the Ushio source several years ago: “the Ushio source [had] degradation and they took it out...the same people who came to me and said, wow, we won the war and we got Lasertec...it’s all good with Ushio...the same people two years later said yeah by the way everybody in the field has taken the source out.” He expressed similar doubt about the Isteq source: “Isteq has not quite done their job”; “Lasertec and Intel will...do a song and dance...well, how much data do you have...I have not seen long-term performance data at all.”

Detailed summary / key findings

4. The “new” ACTIS A300, announced with fanfare in Nov 2023 as Lasertec’s next-generation EUV product and driver of growth, is a hoax (cont’d)

- The expert stated that an advisor to Lasertec confirmed to him that “Ushio’s source has been taken out everywhere.” He eviscerated Isteq’s source as an old, failed technology no one wanted before: “it didn’t use to work well before...so, I’m wondering what happened that all of a sudden this thing is working...this was an LPP source that nobody cared about...what happened in the last couple of years that all of a sudden they have improved their source?...Isteq’s source is a rotating tin target...you’re doing to drip the tin off it easily...it’s not a good situation.”
- An ex-KLA executive in a leadership role in their EUV mask inspection group corroborated the damning color on Isteq’s source. He stated he has visited Isteq and is familiar with the management team, dismissing it as a laboratory R&D project not suitable for commercial use: the “Isteq source will not be done in a professional way and it cannot work...you need to understand Isteq was done for research...research is done in bursts, not continuously...for inspection, you need something to work 24/7...it’s really hard...it’s totally different.”
- In addition to promoting the “new” and supposedly improved light source in the A300, Lasertec purports that the tool is High NA compatible, which refers to next generation EUV lithography with high numerical aperture (“NA”) to shrink the feature size. High NA is similar to regular EUV lithography but with larger optics with a 0.55 NA mirror vs. 0.33 currently, for better resolution and higher wafer printing density, similar to changing the F-stop or aperture on a camera. ASML’s new High NA EUV scanners shrink chip features by 1.7X and triple the transistor density.
- However, changing the aperture to 0.55 for High NA requires completely overhauling the EUV source, optics, and other components required for actinic mask inspection – none of which Lasertec is capable of or appears to have done. In particular, High NA requires an anamorphic lens system and a significantly larger illumination system and projection optics. The German company Zeiss is the only developer of High NA anamorphic optics, as part of a strategic partnership with ASML, which precludes Lasertec from the intellectual property which is “secured by 2,000 Zeiss patents.” Zeiss states that its new High NA Mirror alone is twice as large and ten times as heavy as previous EUV mirrors, and required 1,500 dedicated staff - out of 7,500 total - to develop; and that each mirror takes an entire year to produce as they are “manufactured to atomic precision and consisting of more than a hundred layers...the mirrors are at the heart of High-NA-EUV technology” – impossible for Lasertec to replicate.

Detailed summary / key findings

4. The “new” ACTIS A300, announced with fanfare in Nov 2023 as Lasertec’s next-generation EUV product and driver of growth, is a hoax (cont’d)

- Multiple industry executives in touch with Lasertec told us that the company’s claims around the “new” ACTIS A300 and its purported High-NA capability are simply false and just vaporware – in contrast to KLA’s upcoming machine which is “already ready for High NA.” An ex-KLA executive in a leadership role in their EUV mask inspection group stated that he couldn’t disclose details of Lasertec’s technology given confidentiality agreements, but bluntly indicated that they do not have High-NA: “they don’t have it now, without giving too much info on that – they don’t have it...it will require replacement of their optics because...if they open up the aperture, the optics will block....they cannot do it right now...they need...a different curvature, different figure elements – it’s a massive upgrade.”
- A longtime executive at Global Foundries, who deals directly with Lasertec and all aspects of EUV mask inspection, stated that Lasertec’s November 2023 press release announcing the ACTIS A300 was “just a press release to match the shipment of the first [ASML] High-NA system to Intel.” He added that he has “not even heard of a single A300 system, a 300 system that shipped even...I don’t know if it’s pure vaporware.”
- As a major red flag regarding the A300’s High NA capability, we note a key paper presented by Lasertec engineers a few days before the press release. Presented at an industry conference, the 11-page paper shows the emperor has no clothes, as it was curiously vague on technical details in comparison to ASML’s detailed presentations and contains only one short, 6-line paragraph on the projection optics key to High NA capability – without explaining how their optics could enable inspection of High NA anamorphic patterns which require different magnification in the X and Y directions. We are troubled that the paper is conceptual and written in the future tense as only a vision - versus the press release days later that represents the High NA system as here and now.
- A diagram of the supposedly High NA optics in the A300 - shown in both the conference paper – is astonishingly lacking in technical specs or details. Instead of being transparent and showing photos of their High NA projection optics as ASML and its strategic partner Zeiss have done for years, Lasertec’s engineers show only a simple, crude drawing to compare the optics design in the A150 with the A300. As a red flag, the diagram states the NA – numerical aperture – for the A150 but omits it entirely for the A300 and says only that it is a “larger objective NA.” In addition, the paper resorts to a mathematical model versus actual results from using the tool, repeatedly using the word “simulation” or its variant: “imaging simulations using a rigorous optical simulator...in this simulation, A3XX shows a significant improvement in contrast....”

Detailed summary / key findings

5. Lasertec's EUV ambitions have always been vaporous given the impossibility of developing an EUV light source without investing USD billions. It is not - and will never be - a serious, credible player in EUV lithography/inspection, and investors who perceive it as "Japan's ASML" are misguided. We spoke with two of the most knowledgeable EUV source experts in the world: a leading figure in ASML/Cymer's source development, and a senior R&D individual at Isteq. They elaborated on the history and origin of ASML's source; and Lasertec's struggles and failures with 2 different source suppliers, Ushio and Isteq. Even Isteq stated that the "new" source technology they licensed to Lasertec for its A300 "Urashima" is a pipe dream that is as fatally flawed as Ushio's; that Lasertec's EUV tools are a failure; and that key customers like TSMC and Intel are angry.

- As part of our investigation, we spoke to two of the most important and knowledgeable authorities on EUV light source technology for lithography and mask inspection: 1) Cymer, which developed the first working source and was acquired by ASML for \$4B and became the foundational technology for ASML's EUV scanners; and 2) Isteq, a Netherlands-based affiliate of a Russian institute focused on plasma research and light source development, from whom Lasertec licensed their "new" EUV source, which they renamed "Urashima."
- The Cymer expert spent over a decade working on the EUV light source prior to its acquisition by ASML in 2013 and is one of the top scientists in the field, recently leading the EUV source development program at one of the largest semicap equipment companies, which we don't name to respect his confidentiality. He emphasized the extreme difficulty and cost of developing a reliable EUV source, highlighting the \$4B that ASML spent to acquire Cymer in 2013 – the cost of entry just to have a light source to power their scanners. He stated that Cymer's source still wasn't working in 2013 and that the "scope and size of the investment" that was still required "was way above what Cymer could afford to make EUV technology happen" - "that's why ASML had to buy Cymer."
- The Cymer scientist stated that it is impossible for Lasertec to replicate a tin plasma-based LPP source similar to ASML's, as it claims with its new "Urashima" technology. He stressed that it took Cymer and then ASML a combined 20 years of R&D to make it work, with Cymer having 300 dedicated staff just for the source; that ASML then spent "several billion dollars" more in R&D "on the source only," after acquiring Cymer; and that there are currently ~1,300 ASML staff in a San Diego facility where Cymer was based, dedicated just to the EUV source – "that's the scale"; "Lasertec is not in a position to throw 100 people working on a source."

Detailed summary / key findings

5. Lasertec's EUV ambitions have always been vaporous given the impossibility of developing an EUV light source without investing USD billions (cont'd)

- In comparison to the 1,300 staff at ASML's EUV source group and the \$4B it spent on Cymer plus billions more in additional R&D, Lasertec is small with only 859 total employees in 2023, and historically invested little in R&D – generally \$9MM to \$15MM in R&D in the years preceding its EUV market entry in 2019. It strains credibility that Lasertec was able to develop a reliable EUV source with less than 1% of what Cymer/ASML spent. Ushio – its EUV light source supplier – spends \$80-90MM annually on R&D, but EUV sources are a small percentage of its revenue and EUV R&D expenditure would have been a tiny fraction of the total. Isteq, which developed Lasertec's "new" Urashima EUV source is little more than a Russia/Netherlands-based science project with only 12 employees on LinkedIn.
- The ex-Cymer scientist stated that ASML initially began with Ushio's source in 2009 but that it was a total failure: "it didn't work...they screwed it up"; the Ushio source "has a lot of problems" such as tin-based "discharge that...creates a lot of junk...nearby optics get literally covered in tin pretty quickly" – "still way more dirty than everybody wants it." He detailed the difficulty of developing a working source through ASML's failures for a decade, even years after spending billions on Cymer – illustrating Lasertec's insurmountable EUV light source problem.
- The Cymer expert also slammed the "new" Urashima source that Lasertec has licensed from Isteq, and we view his color as particularly damning given his intimate knowledge of Isteq and its technology. He stated that neither Lasertec nor Isteq has ever shown this source works and that it's just a little lab experiment: "they did not show anything – I am really skeptical"; "as is, it would not work...it's nothing but a lab prototype, slightly more than an idea."
- The expert detailed Lasertec's history of struggling with different light source suppliers and repeatedly failing, stating that they failed to scale from the simple Xenon-based EUV source they procured from Energetiq for their blank inspection tool, as actinic mask inspection is more complex: "the optics are much more complicated and they need significantly more light...that's why they are struggling...they moved to Ushio...they're still struggling...they have to move to something else." He stated that Lasertec "cannot deliver" on their purported \$2 billion backlog due to their lack of a properly functioning EUV light source: "they have a \$2 billion backlog, which means that they probably have orders for 40 or 50 systems that they cannot deliver right now."

Detailed summary / key findings

5. Lasertec's EUV ambitions have always been vaporous given the impossibility of developing an EUV light source without investing USD billions (cont'd)

- The Cymer scientist's assessment of Lasertec's defective light source was corroborated by Isteq, the Russia/Netherlands company that developed the new "Urashima" source that Lasertec is now using. We spoke with a longtime, senior member of the technical team at Isteq, which is a Netherlands entity affiliated with a Russian firm which calls itself "RnD-ISAN" or "EUV Labs." Despite positioning itself as a non-Russian company given the current Russia sanctions, Isteq appears to be a subsidiary of RnD-ISAN. We note that the enterprise is extremely small and appears to be little more than a research institute, showing only 12 staff on LinkedIn.
- The Isteq engineer stated that "we developed it," referring to Lasertec's new LPP source and revealed that they actually licensed it to Lasertec many years ago, making it far from "new": "actually it was quite a long time ago...more than two or three years ago." He emphasized that the EUV source is the heart of the entire machine: "the source part, the lamp, is crucial...absolutely – the source is the key."
- Shockingly, the Isteq engineer stated that the Urashima source they licensed to Lasertec has the same problems as Ushio's: "I don't see why it should be any different...no ultimate solution...there is no cure" – echoing the Cymer scientist's conclusion that Lasertec's tin-based approach is doomed no matter what. Isteq detailed how Urashima produces the same contamination as Ushio: "LPP machines are susceptible to different types of contamination...millions of micro-explosions...a lot of debris and vapor and tiny particles which have high speed and energy...like bullets...problem of almost all LPP laser-produced plasma machines...Lasertec's source...it's definitely not an exception...tin vapor everywhere...they ricochet...and they penetrate almost everything."
- Isteq bluntly dismissed Lasertec's Ushio source and its DPP discharge-based technology as a complete failure: "DPP...machines have no future...Ushio made a strategic mistake because they were somewhat deluded by the deceptive simplicity of DPP and they chose this way mistakenly." He added that Lasertec's problems go far beyond the source and encompass other components: "the inspection machines are not in a mature state...this is the main problem in the industry...most of the designs...are plagued by different problems"; Lasertec's "technical problems stem from two different sources...the [light] source and the second is...the testing part of the system itself."

Detailed summary / key findings

5. Lasertec's EUV ambitions have always been vaporous given the impossibility of developing an EUV light source without investing USD billions (cont'd)

- Isteq indicated that they are in touch with Lasertec's key customers – namely Intel and TSMC - and that they are unhappy with Lasertec's inspection machine: “it's under NDA so I cannot reveal the nature” of the communication; “no company...is fully satisfied with their inspection product...absolutely every product now has different problems...”APMI, actinic patterned mask inspection is a huge challenge.”
- The Isteq LPP laser-produced plasma technology upon which the “new” Urashima source is based was developed over a decade ago. Lasertec could easily have licensed it prior to the ACTIS A150 launch in 2019 but chose Ushio instead. Given that there are only two or three EUV source suppliers to choose from, it is obvious that Lasertec would have evaluated Isteq over five years ago prior to rejecting it for Ushio – and are now desperately trying to cover up the Ushio failure and trying Isteq as a Hail Mary. We note various Isteq papers going back to 2013 on their LPP system; as well as an international patent application for their LPP technology back in 2019/2020,

Detailed summary / key findings

6. Lasertec's largest customers – notably TSMC, Intel, and Samsung, which are 77% of its revenue – are struggling, exasperated, and/or furious with its EUV mask inspection system, as well as its DUV systems, which we believe to still comprise the vast majority of its revenue. Key customers are “really struggling about what to do”; “it’s painful, extremely painful” to use Lasertec’s tool; “it would be stupid” to buy more machines.

- An ex-KLA executive, who was in a leadership role in their EUV mask inspection group, described his direct conversations with Lasertec. He stated that he began discussions “two years ago” with key customers who are “really struggling about what to do” with Lasertec: “the discussions are really hard.”

While Lasertec’s purported “monopoly” position is dependent on inspecting EUV masks with a pellicle – a thin membrane to protect the mask from debris – it cannot reliably do so and the problems with its tool are “too hard to fix and it takes too much time.” Customers who bought it hoping to inspect with pellicles “have been paying a huge price” and now simply take the risk of inspecting without a pellicle rather than deal with the tool: “it’s painful, it’s extremely painful”; “they basically print without a pellicle”; “all these failures...slows down all the flow in the fab”; “it would be stupid to buy many more” machines; “I would do everything I can to avoid it.”

- Customers are furious because they buy Lasertec’s tool in order to inspect masks with a pellicle, but then can’t use a pellicle given its inspection deficiencies, which compromises their entire inspection workflow given the risk of debris from not using a pellicle, resulting in yield issues, delays, and bottlenecks: “it’s really a huge issue for the fabs”; “if you don’t have a good actinic tool, you cannot use a pellicle”; “a particle will fall on the mask...every wafer will be zero yield”; “so what is the benefit?...don’t use a pellicle but it’s very painful.” In the absence of pellicle inspection capability, customers just default to their legacy KLA mask inspection tools: “because you don’t have a pellicle...so you just use the KLA tool” – negating the “EUV monopoly” thesis underpinning Lasertec’s stock.
- Lasertec’s customers can’t reliably use it for P3 inspection, thus undermining the main reason for buying it and providing no benefit versus KLA’s legacy DUV inspection tool that dominates P1 inspection. We note that P0 refers to inspection in the mask blank shop, done by an inexpensive tool; P1 is after the blank is printed with a mask pattern, and is dominated by KLA’s Teron DUV line; P2 refers to re-inspecting the mask after repairing defect; and P3 is after adding a pellicle. Lasertec’s only use case is P2/P3, but it introduces debris in P2 which requires redundant inspection with a backup KLA DUV tool, and it’s too unreliable for P3 – eliminating any reason for its existence versus KLA.

Detailed summary / key findings

6. Lasertec's largest customers – notably TSMC, Intel, and Samsung, which are 77% of its revenue – are struggling, exasperated, and/or furious (cont'd)

- Due to “the struggle of the Lasertec tool both on the performance side and...on the stability or uptime side,” key customers are “begging” KLA to hurry their upcoming actinic mask inspection tool: “Oh my God...the amount of pressure, it’s inconceivable...the pressure is insane...from Samsung and TSMC combined...TSMC number one.” Customers complain that its lack of reliability requires them to purchase redundant tools: “that’s why sometimes there are two tools...they need to have a backup tool...it breaks everything...and that’s the number one problem they have.”
- We asked if Intel and TSMC “hate” the Lasertec tool, and the expert replied “correct” and stated that their dilemma is to either not use a pellicle and avoid using Lasertec - TSMC’s approach - or to use a pellicle and suffer with the lack of reliability until KLA introduces its alternative. He stated that during his role in the KLA’s EUV mask inspection group, key Lasertec customers were coming “crying” to KLA and “begging us to expedite”: “I’m talking about customers coming to KLA... saying, guys, we need you to develop this tool...P2 can only be done by a clean tool...they were really begging us to expedite.”

Detailed summary / key findings

6A. Struggles at Intel. Lasertec's EUV mask inspection effort is Intel's baby, which it sponsored with \$300-500MM in advances >5 years ago as its "founding" customer after failing to convince KLA. Intel's mask group is trying to save face and cover up that its investment has been a debacle, giving it supplier prizes for public PR while complaining behind the scenes, according to multiple research sources. A Lasertec field engineer – based on-site at their Santa Clara, CA mask shop – revealed that Intel's mask shop head was so enraged with the tool's failure that he was banned from attending meetings for blowing up at staff. One of Intel's most senior and tenured executives, with a leadership role in EUV, confirmed that Lasertec's EUV source "is not a well-controlled process...it was painful...if you choose a bad technology you just can't get there"; Intel's VP mask operations VP confirmed the struggles.

- Lasertec - in its current incarnation as an "EUV inspection equipment" stock – is Intel's baby. Circa 2018, the industry began planning to use ASML's EUV scanners for 7nm and 5nm production. However, the lack of actinic patterned mask inspection (APMI) – inspecting EUV masks through a thin protective membrane called a pellicle – became a major gap in the EUV lithography supply chain. Our sources indicate that Intel pushed KLA to develop an EUV mask inspection tool, but KLA asked for too much money. Intel then pressured Lasertec to rush a tool to market. A 2018 trade publication stated that Lasertec "obtained funding for the technology" from "a large chipmaker" – asked if this was true, Lasertec replied "no comment."
- Research sources told us that Intel invested \$300 to \$500 million in Lasertec a few years ago via advances for future purchases but "at a significantly reduced price...they don't have a lot of margin on those tools." A longtime industry analyst who recently completed a consulting project for Lasertec and remains in touch with management stated that Intel "do have a significant say in the direction of Lasertec, and Lasertec views them as sort of their core customer or founding customer...they've obviously ordered and prepaid...."
- Multiple semiconductor executives confirmed that Intel is just as dismayed as TSMC and is similarly struggling with Lasertec's EUV tool. A longtime semiconductor equipment friendly with Lasertec's CEO stated that "I have met the VP of Intel's mask operations, a few times." He indicated "they are very secretive" and replied "yes, of course" when we asked if Intel is having the same problems as TSMC. The ex-KLA EUV mask inspection executive corroborated that Intel is complaining behind the scenes – without naming them specifically but leaving no doubt as to which "major US customer" he was referring to.

Detailed summary / key findings

6A. Struggles at Intel (cont'd)

- We tried speaking with numerous Intel employees in its EUV mask inspection effort. One of Intel's most senior and tenured individuals, with a leadership role in EUV and the Intel/Lasertec relationship, agreed to a phone consultation but only at 300% of the usual expert rate – and then played games and revealed little. Intel's mask team clearly realized someone is investigating Lasertec and agreed to calls only to try and deduce our identity – another senior individual, from its mask shop and who we think spoke to the first executive, agreed to a call and then demanded an absurd \$5,000 expert fee immediately prior to the scheduled time (we declined). Nonetheless, the senior executive with whom we spoke did make a few damning admissions, saying that Lasertec's light source is “not a well-controlled process...it was painful but we would get some light...if you choose a bad technology you just can't just get there...DPP has inherent issues.”
- We first realized that something is amiss as we read Intel's EUV mask inspection presentations at SPIE, the main industry association. We noted three presentations by Safak Sayan, a ~20-year Intel veteran per his LinkedIn profile and who we believe to be the head of Intel's mask shop. The presentations are typically co-authored with Lasertec employees, and seem rather eager to convey that the tools Intel purchased actually do work – but exhibit obvious red flags indicating the opposite. A key chart for the tool's uptime omits the Y-axis units entirely - a cliché case of “How To Lie With Statistics.” Another paper suggests ongoing struggles, and once again, key performance charts lack credibility.
- In glimpses of candor, the Intel SPIE papers nonetheless confirm challenges with Lasertec's EUV tool, namely contamination and optics degradation from tin plasma (“ion induced damage to the optical components”); light source stability and brightness level; struggles in managing source brightness vs. source chamber pressure, as high pressure is needed to help mitigate tin debris but which reduces brightness; and unstable plasma which leads to dark pulses and inspection failures – “the plasma becomes unstable and brightness stability becomes worse...brightness may drop below the threshold required for mask inspection.”
- An ex-Lasertec field service engineer provided us with particularly astonishing color on Intel's actual, behind-the-scenes view of Lasertec. The engineer was until recently based full-time, on-site at Intel's mask shop as part of nine field engineers servicing three Lasertec EUV tools. He was embedded with Intel staff and essentially their colleague. He stated “there was definitely a lot of frustration around” Lasertec's system, and that Intel's head of EUV mask inspection was so livid that “he got banned from a couple of meetings for blowing up on people” – we believe this to be the same person presenting the SPIE papers.

Detailed summary / key findings

6A. Struggles at Intel (cont'd)

- When we asked the Lasertec engineer for more detail on the tension, he stated that the head of Intel's EUV mask inspection group was responsible for the tools – "the tools were his main area of management at Intel" – and that he was so angry that "he basically wasn't allowed in certain meetings anymore" by "his higher-ups": "he wanted utilization closer to 85-90%, and being around 60-65% was not cutting it for him...he just tried to press, press, press to make it happen but there was only so much that could be done." This, of course, explains the statistical omissions in the SPIE papers lauding the tool's purported uptime.
- At the outset of our research, we were stumped as to why Intel includes Lasertec in its annual list of supplier awards – until we understood Intel's role as Lasertec's sponsor and anchor investor, and the natural incentive of its EUV mask inspection team to avoid embarrassment, particularly given Intel's interest in promoting its EUV progress to investors and players like Apple. When we asked a senior Marvell employee about the award, he mocked it and laughed – "Marvell has also received that award...[it] is really sitting inside a cubicle that nobody uses...it's not something that we write home about...yeah we'll do a press release...but nobody talks about it"; "Intel is not a cutting-edge company in this day and age by any means...so if Intel is giving out awards...it's not even worth its weight in salt."

Detailed summary / key findings

6B. Struggles at TSMC. The world's largest chipmaker's experience with Lasertec has been a disaster, not only with its ACTIS EUV systems but also with its outdated, previous-generation DUV tools. Multiple sources including TSMC's ex-head of R&D, a procurement executive who first brought Lasertec into TSMC, and the head of TSMC's mask shop – one of whom even laughed at Lasertec's tool - indicated that TSMC stopped buying ACTIS machines years ago; that they are so unreliable that masks require parallel, redundant inspections with older tools; that at best it is a niche, laboratory R&D tool unsuitable for fabs; that KLA's DUV tools are superior for EUV mask inspection and that KLA has already taken 2/3 of their mask inspection share. Lasertec's status is so dire that TSMC has forced them to attend a standing, weekly crisis meeting: "it's just not stable...it doesn't work...it just dies"; "TSMC is not happy with Lasertec"; "TSMC will not consider the A300 or A150 for production." Critically, four different TSMC sources indicated to us that the "new" ACTIS A300 with the Urashima light source recently failed their evaluation process.

- Multiple research sources indicated that TSMC's experience with Lasertec has been a disaster, not only with its ACTIS EUV mask inspection line but also with its outdated, previous-generation DUV MATRICS X8/X9 machines. We spoke with TSMC's former head of TSMC R&D, who played a key role in its 5nm transition and vendor/tool selection; a former longtime TSMC procurement executive, who stated that he originally introduced Lasertec's tools into TSMC and is in touch with current TSMC executives as well as with Lasertec management; a senior semicap executive in Japan who is friendly with Lasertec's CEO; and an ex-KLA executive in a key role in their EUV inspection group, who is also in contact with Lasertec.
- The ex-head of TSMC R&D stated that they still rely mostly on traditional KLA tools, not Lasertec actinic inspection: "Taiwan is very smart. They always wait for Intel and these guys to go first...they usually do not want to take the risk of being the first...that happened for EUV...were really slow...still used the traditional tools like KLA" which "has very strong inspection mask manufacturing for EUV."
- A longtime semicap executive in Japan - who speaks with Lasertec's CEO – said that he was "in charge of TSMC for about 15 years" at another tool supplier and "I was dealing with all the TSMC management people in the mask shop. I know how they function." He confirmed the color from TSMC's ex-head of R&D that TSMC can't rely on Lasertec's tool. In particular, he stated that he recently met the head of TSMC's EUV mask shop - whose name we redact – and that he evaded questions about their struggles with Lasertec: "If they didn't have any [issues], he would happily talk about how things are going." The expert conveyed that he knows the head of TSMC's mask shop well.

Detailed summary / key findings

6B. Struggles at TSMC (cont'd)

- The executive in Japan confirmed TSMC's desire to cover up its problems with Lasertec, after we asked if a 2023 SPIE paper co-authored by Lasertec and TSMC was an attempt by mask shop employees to save face after spending millions on tool purchases: “they placed an order, I think, for 10 tools, so they cannot come out with any negative papers [about Lasertec]”; “even upper management who accepted the purchase order will be put in an awkward situation...TSMC is a very hierarchical company...you don't want to make any superior people look bad.” We note that the TSMC mask shop executive that the expert spoke with is listed as one of the authors in the paper, shown below.
- The semicap executive was skeptical as “to what degree [TSMC] have succeeded in phasing in Lasertec tools to the production cycle” – confirming issues in both of its use cases in the mask shop as well as the fab.” Because TSMC is failing to get results from the ACTIS machine, he stated that they have to rely on redundant, backup inspection with traditional KLA and other tools: “so they need a combination of optical DUV and electronic beam inspection to inspect the mask.” He cited strong, recent adoption of a Holon e-beam tool by Intel, Samsung, and TSMC as evidence that they are struggling with Lasertec and resorting to backup solutions: “TSMC was the first one to adopt the Holon tool...other customers in the EUV space are also adopting” it.
- The executive stated that TSMC and “pretty much all customers are struggling in phasing in the Lasertec tools” due to “uptime, light source problems” and that “customers a) haven't been successful with the qualification and b) the demand is low.” As a result, he indicated that TSMC is delaying Lasertec orders and that current sales are “probably pre-orders or initial orders” from several years ago.
- Echoing a theme we detail in a separate section, the expert stated that Lasertec's tool is so unreliable and has such little trust that TSMC has to run parallel, duplicate inspections with other tools to catch mask defects: “not so easy with a new technology like actinic inspection...they will have to run parallel inspections of the same masks...just to make sure there are no defects that go through without detection...if it's a fatal defect, it can cost them billions...they cannot ship wafers anymore.”
- Some of the most damning color we received about TSMC's experience with Lasertec's tools is from a former longtime TSMC procurement executive, who stated that he originally introduced Lasertec's tools into TSMC and is in touch with current TSMC executives as well as with Lasertec management. He slammed the ACTIS A150's reliability: “the big thing I hear offline is it's not very reliable” and “needs a lot of support and it has a lot of issues.” Referring to the “new” and purported A300, he stated bluntly that “TSMC is not buying the High NA Lasertec tool...there's no confidence in my opinion that it will be a reliable product.”

Detailed summary / key findings

6B. Struggles at TSMC (cont'd)

- He detailed TSMC's frustration and exasperation with Lasertec around a variety of issues, such as pitifully low mask inspection throughput and problems in inspecting with a pellicle, and described it as "kind of an endless negotiation...to try and convince Lasertec to improve their performance...to try to convince them to change their performance, the quality." He indicated that TSMC has no real interest in purchasing from Lasertec, and that both TSMC and Intel do so only to have pricing leverage with KLA: "TSMC will keep a minimal share for Lasertec...we believe [Intel] purchased Lasertec because they want to have some leverage power with KLA because KLA's price is even higher...TSMC spent a lot of money on KLA's particle inspection tools...that's very, very expensive."
- The TSMC executive indicated that KLA's legacy DUV tools have superior economics, productivity, and throughput versus Lasertec's EUV systems, and that "KLA's share continues to increase" at TSMC: "it's already two-thirds, and I know it continues to increase"; "already two-thirds migrated from originally Lasertec to KLA...I have to say Lasertec has continued to decrease their share in TSMC." In particular, he criticized Lasertec's EUV source: "the big thing is the light...for production, for productivity, and for economic considerations, KLA performs better than Lasertec in TSMC."
- The TSMC executive said that he recently spoke with a senior individual at Lasertec, who stated that things are "not well." The executive stated that TSMC views Lasertec as merely a niche R&D tool that is not suitable in a fab: "we call it a lab tool, R&D tool...not a production tool...because you have a lot of limitations for productivity." We did a second call with the executive after he checked with current staff at TSMC for any change in their negative opinion of Lasertec, and he indicated the situation is dire enough that TSMC holds a weekly meeting in Taiwan with Lasertec: "my colleagues in procurement and from the fab told me they have a weekly meeting to push Lasertec to improve...to improve their performance...their productivity." In contrast, he stated that "KLA improved a lot" with their legacy DUV systems which they rely instead for EUV: "they have a very quick scan, and now their resolution is getting much, much better."
- He stated that TSMC views the Lasertec ACTIS EUV tool as just a niche R&D tool for their mask shop; that it is unusable for production in TSMC's fab; and that TSMC has stopped buying any more of Lasertec's flagship ACTIS A150 machines, replying "no, unfortunately no" when we asked if they're still willing to buy them. He emphasized that "it's not a tool to support production" and that "the Lasertec tool cannot meet demands in a production line."

Detailed summary / key findings

6B. Struggles at TSMC (cont'd)

- He indicated that TSMC only purchased six of the ACTIS EUV machines at \$50 million each, mostly around 2019/2020 and don't appear to have purchased any more in years: “the pricing is so expensive.” TSMC's only use case is in the mask shop, which requires only a small number of machines – we note that Intel's mask shop only appears to have three Lasertec machines. Hence, the TAM for EUV mask inspection machines is entirely dependent on their use in a production environment in a fab, where TSMC stated it is unusable: “it's not for production, it's for EPL,” referring to their internal name for their mask shop.
- The TSMC executive repeatedly and strongly emphasized that Lasertec's tool is a failure that cannot be used inside the fab to inspect masks during production, given its low throughput and inability to inspect more than a few masks per day: “No. TSMC is not happy with Lasertec...it's very expensive but the throughput is slow.” He slammed both the current ACTIS A150 and the purportedly “new” A300 with the Urashima light source: “because of the throughput, TSMC will not consider the A300 or A150 for production...they will only exist in the...mask house.”
- The Lasertec machine's throughput is so low that it can only inspect “five pieces per day” – referring to masks – while TSMC's 5nm Fab 18 needs 550 masks inspected daily in production. Of the six ACTIS EUV tools TSMC has purchased, only two are in the fab and the rest are in the mask shop: “the problem is the throughput...they only have two in the production area because the throughput is too slow...it's not affordable for production demands...it's not economical.”
- The TSMC executive indicated that the “new” ACTIS A300 with the Urashima light doesn't work and that it failed TSMC's evaluation: “Yes. It failed” – laughing as he told us.” He indicated that the evaluation occurred last year: “TSMC will not consider the A300 or A150 for production.”
- Other experts in touch with TSMC and Lasertec confirmed that TSMC is highly disgruntled with Lasertec's EUV tool: “what I know is that TSMC is upset because the source is not working...TSMC is upset...the source does not have uptime, cost of service is too high, they are flickering.” He corroborated that TSMC evaluated the A300 and that it failed and that Lasertec has to write off its inventory: “Lasertec probably must write off those tools.” The expert stated that TSMC began pushing back Lasertec EUV orders over a year ago, after the new light source failed inspection in early 2023 after six months of evaluation: “it's not stable...it's just not stable...it doesn't work all the time...it just dies.” He replied affirmatively when we asked if TSMC is angry with Lasertec.

Detailed summary / key findings

6B. Struggles at TSMC (cont'd)

- TSMC has concluded that EUV mask inspection is a failure given Lasertec's defective technology, and now simply uses legacy DUV tools instead with a ~200nm light source – stating that DUV is sufficient for EUV masks in production. As a result, TSMC's purchases are focused on KLA and Lasertec's old DUV tools called MATRICS X8 and X9 – not its ACTIS EUV line. The TSMC executive stated that “if you care about production throughput, you have to use a 213nm light source.” He stated that even though EUV is 13.5nm and can theoretically catch smaller defects, DUV is sufficient for the quick mask checks that TSMC needs to do in production: “TSMC just wants to do quite frequent monitoring...a quick check...the best resolution is not necessary.”
- However, although TSMC is forced to rely on Lasertec's old DUV tools versus EUV, its DUV line – known as MATRICS X8 ULTRA, and the more recent X9 – is plagued by light source defects as serious as those in its EUV source, in addition to other problems. An ex-Lasertec field service engineer who was recently based at TSMC's fab in Taiwan, whose main responsibility was the MATRICS DUV machines, provided color as damning as that provided by other field engineers for the ACTIS EUV line: “the laser source is unstable for the X8, X9 ULTRA”; “it has a lot of software bugs”; “bug is always causing random problems...they cannot find critical defects, or the resolution is very bad...the image processing is very bad.” Although TSMC can get by with its current DUV machines, the engineer stated that Lasertec's product simply won't work for TSCM's 3nm node.
- The Lasertec field engineer stated that the MATRICES DUV product suffers from particle contamination issues just like the ACTIS EUV line, leading TSMC engineers to complain: “...design is very bad because of the particle issue...the Lasertec machine isn't designed very well...it's suffering from so many particle issues...[the tool] is not very clean.” The tools require a shocking 10 hours of daily cleaning to address particle contamination, leading to a low uptime of less than 60%, with an onerous protocol including UV light to scan for stray particles.
- The field engineer stated that Lasertec's DUV light source is unstable for not just the old MATRCIS X8 DUV product but for the new X9 as well, which was an attempt to address the X8's source problems – stating that he heard this a few months ago “from my colleagues who are X9 ULTRA team members” at TSMC's fab in Taiwan. The similarity between both the light source and particle contamination issues for Lasertec's DUV and EUV products suggests a fatally flawed and defective technology platform.

Detailed summary / key findings

6C. Struggles at other customers, such as Samsung, Marvell, and Global Foundries, which even laughed at the quality of the Lasertec's EUV tool and ceased new purchases in 2021, calling it “just unacceptable” with a “problematic light source” and “not as sophisticated or as capable as the classic [DUV] KLA mask inspection tools; mocked their A300 announcement by saying “no question...it sounds sketchy”; and confirmed that other customers are equally “frustrated and disappointed” and are refusing to take shipment. As a sign of Lasertec's urgent situation, they stated that “senior technical people” from Intel and Samsung recently flew to Japan to express their “concerns with the equipment.” Marvell stated it also has no interest in further ACTIS purchases; that ASML “echoed the same problems that we are seeing”; expressed skepticism of the A300; revealed that Lasertec offered them ~75% discounts in a desperate attempt to offload inventory; and that they recently saw high levels of inventory piling up at Lasertec, leading them to question their inventory valuation and accounting.

- Global Foundries, one of the largest semiconductor companies in the world, blasted Lasertec's EUV product, corroborating the damning color from TSMC and Intel with regards to a defective light source, contamination, and unacceptably low uptime. We spoke with a longtime executive who has been directly involved in purchasing Lasertec systems and in all aspects of mask inspection. He stated that the uptime is only “around 60-70%...disappointing”; “the uptime is clearly an issue...not only uptime but also debris from the tin-based system that they use.” He summarized the issues as a “problematic light source, uptime, and user interface not as friendly or capable as a comparable KLA system.” He even described their “new” Urashima light source as “the risk for them, no question...it sounds sketchy.”
- Global Foundries actually laughed at the quality of Lasertec's tool during our research interview, describing it as merely a “niche player” that has “struggled” with low uptime and a poor user interface, stating that it is highly inferior to KLA. The executive slammed its uptime as “just unacceptable” for inspection equipment: “doesn't meet what we would normally expect”; “if it was not the only actinic inspection tool in the market, people would definitely go with another supplier”; “struggled with uptime concerns”; “not as sophisticated or as capable as the classic KLA mask inspection tools.”
- The executive indicated that Global Foundries only purchased two of the Lasertec EUV actinic inspection machines and haven't bought any since 2021, disparaging it exactly as TSMC did in our interviews as merely an “occasional R&D tool” that can't be used in a fab/production environment.

Detailed summary / key findings

6C. Struggles at other customers, such as Samsung, Marvell, and Global Foundries (cont'd)

- The executive stated that Lasertec's customers are now delaying "shipment of systems" due to the rampant product problems and it's product "not even being adequate": "they're holding off near-term shipments until they resolve these quality issues." He indicated that its defective EUV system has badly damaged Lasertec's reputation and described its customers as "frustrated and disappointed": "prior to EUV, Lasertec was known as a reasonable supplier"; "[customers] are frustrated...frustrates people because their brand has definitely taken a hit as a result of their EUV inspection equipment"; in the past Lasertec "was adequate, and that's the frustration today – not even being adequate."
- The Global Foundries executive stated that "senior technical people" from Intel and Samsung - peers who he spoke with - recently visited Lasertec's headquarters to convey concern about its product, indicating the severity of frustration among all three of its top customers: "they were visiting in part because of...quality concerns with the equipment, and to get a first-hand view of what is done in the factory to improve outgoing quality": "it was more than somebody just going there to complain...these are senior technical people."
- Marvell, one of the largest semiconductor companies in the world, expressed the same dismay with Lasertec's EUV tool as TSMC, Intel, Samsung, and Global Foundries. They stated that they purchased one ACTIS A150 EUV inspection machine and lease two others, with no interest in purchasing any more. Marvell noted the same light source defects and technical problems as other large players. In particular, a Marvell senior employee stated that he spoke with ASML, and that ASML "echoed the same problems that we were seeing...ASML customers were also reporting the same things with Lasertec with respect to the light source."
- Marvell stated that it is "aware" of the "new" ACTIS A300 EUV inspection machine but that "it doesn't make sense for us right now" to even evaluate it. The senior employee expressed significant skepticism: "there was no data that they had"; "no real-world case studies from independently verified semiconductor industry associations"; "no real way to discern that this actually has a value-add."
- The Marvell senior employee stated that one of their executives recently visited Lasertec's headquarters in Japan, and that Lasertec was desperate to offload its ACTIS EUV inventory, offering up to ~75% discounts on the A150, as well as other inducements and concessions that acknowledged uptime and reliability problems: "they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of...we were getting a reduced price as well if we wanted to purchase additional A150's...in the range of \$15-30MM." Marvell stated they previously paid \$62MM for the A150.

Detailed summary / key findings

6C. Struggles at other customers, such as Samsung, Marvell, and Global Foundries (cont'd)

- The details that Marvell provided suggest that Lasertec is desperate to offload bloated EUV inventory: “they made a multitude of offers to try to get us to basically purchase more equipment.” Notably, Lasertec offered terms that indicate an attempt at damage control given customer frustration with product defects and low uptime: “other concessions were made like extended warranty service...at no additional cost...even guaranteeing service time like 60-70%...if not then we could take some kind of reimbursement...for the lost time of use” – making us question how Lasertec could retain any margin given the three field service engineers required to maintain each machine plus the millions to service, clean, and replace the light source and optics.
- Marvell told us that when the senior executive visited Lasertec in December, Lasertec offered an arrangement where “they would actually swap out the A150 that we have with an A300 at a discounted price...because they wanted to try to get early purchase volume.” The offer suggests not only desperation but an attempt to mislead investors by essentially fabricating sales of the “new” A300, which is simply the A150 with a different light source and a new name in an attempt to cover up a defective light source. Marvell indicated that they declined the offer and that “we haven’t seen real proof of concept and enough case studies to actually show the real-world application of it versus the A150.”
- Marvell was one of several customers we interviewed who recently visited Lasertec’s headquarters and reported seeing an unusual level of machine inventory – the Marvell executive “mention[ed] a build-up of inventory when they visited the Lasertec headquarters – and indicated skepticism of Lasertec’s inventory valuation: “I can tell you right off the bat...they haven’t taken any impairments against their balance sheet, so they’re probably over-valuing their inventory far greater than it’s actually worth...a lot of these companies are sitting on so much inventory....”
- Marvell indicated that Lasertec’s existing inventory has little value, given the offer Lasertec recently made to “take back the A150 and then give is an A300 at a reduced price...the value of the A150 that we have on hand wouldn’t be worth that much.” The senior employee noted that inventory mis-valuation in the semiconductor equipment industry “is a common thing, unfortunately, that I’ve seen across multiple companies in the space...to manipulate earnings: “they defer out...that impairment...because they don’t want to record a hit on their gross margin.”

Detailed summary / key findings

7. Lasertec customers who have recently visited Lasertec's headquarters confirmed seeing highly unusual levels of inventory - "pretty scary...pretty weird" - and other red flags: their "factory floor for finished goods is full.. a substantial amount of finished goods inventory that's either not passing [evaluation]" or "customers have deferred shipment"; "literally no space left on their final inspection bays before the equipment ships." They indicated that customers are refusing to accept shipment of tools; that Lasertec "recognized revenue for prepayments of systems that aren't yet shipped"; that most of their initial advances and purchase orders were from Intel and are being "pushed, pushed, pushed out" and will never be filled, with Intel allegedly trying and failing to pawn off PO's to TSMC; are misclassifying inventory using "tricks that any supplier can play"; inflating the value of inventory - "they're going to have to significantly mark down"; "haven't taken any impairments"; "they don't want to record a hit on their gross margin"; and detailed aggressive, questionable discounts, trade-ins, inducements, and concessions Lasertec is quietly offering customers, as well as "reimbursement...for the lost time of use" due to the ACTIS tool's defects and lack of uptime.

- Lasertec customers who recently visited their headquarters confirmed seeing an unusual level of inventory piling up, and speculated that they "recognized revenue for prepayments of systems that aren't yet shipped." A Global Foundries executive conveyed color "from colleagues that I've spoken with" - "their factory floor for finished goods is full...a substantial amount of finished goods inventory that's either not passing [evaluation by customers] - customers have deferred shipment...but from colleagues who have visited, they said they have literally no space left on their final inspection bays before the equipment ships."
- Global Foundries indicated that customers are refusing to accept shipment of the tools: "I've heard of a delay of shipment of systems to have Lasertec fix some things or engineering changes...they're holding off near-term shipments until they resolve these quality issues." They commented further that misclassifying finished goods inventory as work-in-process is easy: "tricks that any supplier can play...you don't have to place it ready for shipment...any number of ways you can say it's work-in-process...if you don't do that final quality inspection....any number of minor steps...you hold off doing them and you still get to call it WIP."
- Global Foundries noted "a huge risk to Lasertec, if machines are piling up and KLA is going around reassuring people that if you can hold off until mid-2025" - a perfect storm where customers are first refusing shipment due to product defects and then simply waiting for KLA's upcoming system - "Lasertec could be caught with inventory they're going to have to significantly mark down...they would end up having to impair...an asset write-down."

Detailed summary / key findings

7. Lasertec customers who have recently visited Lasertec's headquarters confirmed seeing highly unusual levels of inventory (cont'd)

- Marvell also recently visited Lasertec's headquarters and reported seeing an unusual level of machine inventory – the Marvell executive “mention[ed] a build-up of inventory when they visited the Lasertec headquarters” – and indicated skepticism of Lasertec's inventory valuation: “I can tell you right off the bat...they haven't taken any impairments against their balance sheet, so they're probably over-valuing their inventory far greater than it's actually worth...a lot of these companies are sitting on so much inventory....”
- The Marvell executive who recently visited Lasertec's headquarters in Japan indicated that Lasertec was desperate to offload its ACTIS EUV inventory, offering up to ~75% discounts on the A150, as well as other inducements and concessions that acknowledged uptime and reliability problems: “they were trying to also offer additional A150's that they had on the books that weren't, let's say, being taken ownership of...we were getting a reduced price as well if we wanted to purchase additional A150's...in the range of \$15-30MM.” Marvell stated they previously paid \$62MM for the A150.
- The details that Marvell provided suggest that Lasertec is desperate to offload bloated EUV inventory: “they made a multitude of offers to try to get us to basically purchase more equipment.” Notably, Lasertec offered terms that indicate an attempt at damage control given customer frustration with product defects and low uptime: “other concessions were made like extended warranty service...at no additional cost...even guaranteeing service time like 60-70%...if not then we could take some kind of reimbursement...for the lost time of use” – making us question how Lasertec could retain any margin given the three field service engineers required to maintain each machine plus the millions to service, clean, and replace the light source and optics.
- Marvell told us that a senior executive visited Lasertec in December, and that Lasertec offered an arrangement where “they would actually swap out the A150 that we have with an A300 at a discounted price...because they wanted to try to get early purchase volume.” The offer suggests not only desperation but an attempt to mislead investors by essentially fabricating sales of the “new” A300, which is simply the A150 with a different light source and a new name in an attempt to cover up a defective light source. Marvell indicated that they declined the offer and that “we haven't seen real proof of concept and enough case studies to actually show the real-world application of it versus the A150.”

Detailed summary / key findings

7. Lasertec customers who have recently visited Lasertec's headquarters confirmed seeing highly unusual levels of inventory (cont'd)

- Marvell indicated that Lasertec's existing inventory has little value, given the offer Lasertec recently made to "take back the A150 and then give is an A300 at a reduced price...the value of the A150 that we have on hand wouldn't be worth that much." The senior employee noted that inventory mis-valuation in the semiconductor equipment industry "is a common thing, unfortunately, that I've seen across multiple companies in the space...to manipulate earnings: "they defer out...that impairment...because they don't want to record a hit on their gross margin."
- An ex-KLA executive in a senior role in their EUV mask inspection group stated that Lasertec's EUV tool shipments hit the wall about two years ago with delays due to product defects: "I'm telling you, that's where the problem started two years ago and they said that we cannot ship...they are not shipping because it's a liability...I definitely understand why because they are not shipping...because they have problems with the source...the customers are delaying orders."
- The ex-KLA executive described Lasertec's \$1.2 billion of inventory as "pretty scary...pretty weird...doesn't seem normal...that's pretty bad." He stated that most of the initial advances and purchase orders for the EUV tool were from Intel, their "founding customer" – "when it started, most of the slots were Intel slots" – but that they are now being "pushed, pushed, pushed out" and will likely never be filled, with the PO's just sitting on Lasertec's balance sheet as advances until the inevitable impairment charge and capitulation: "Intel...delayed those slots...they will never cancel those PO's...because of the investment they got...it's just there because they owe Intel money..."
- The executive indicated that Intel gave Lasertec larger advances and more purchase orders than it could use and then tried to pawn them off to TSMC. However, TSMC – as detailed in a previous section – is fed up with Lasertec's product defects and has ceased buying their EUV inspection machines – "TSMC is...delaying purchasing until they fix their offload problem" and that they started delaying "a year ago now." He stated that TSMC evaluated the "new" ACTIS A300 with the Urashima light source in early 2023 and that it failed their qualification process.
- Other research sources corroborated the mountain of inventory, such as a senior semiconductor executive in Japan – who stated he is friendly with Lasertec's CEO and that customers are "holding back the orders, waiting for improvement light source-wise or whatever"; "pretty much what the reality looks like...the reason why they are sitting on top of a lot of inventory."

Detailed summary / key findings

8. Lasertec's small, one-time EUV market opportunity is already exhausted, with little to no growth left in its flagship ACTIS line. Aside from product defects, irate customers, and the lack of a viable EUV source, an analysis of its growth drivers in the last 5 years indicates they were ephemeral: a) the industry transition to the 5nm node, which is now complete and during which there was a momentary window for its EUV system; b) the ratio/correlation of EUV mask inspection tools per ASML EUV scanner, a canary in the coal mine as ASML's shipments have hit the wall; and c) the installed base and trend of ACTIS tools inside each use case at TSMC, Intel, and Samsung, whether mask shop or fab. As Intel and TSMC's mask shops only have 5 ACTIS tools in total and ceased new installs, and TSMC views it as a failure in the fab, Lasertec's only remaining hope is Intel adding tools in its fabs – a fantasy, as most of its recent revenue “growth” has come from movements in its deferred revenue and advances account, with therefore little cash flow.

- Product defects, customer frustration, and exploding inventory naturally indicate that Lasertec's small EUV market opportunity is exhausted, with little growth for its ACTIS line. In addition, its growth since the introduction of its EUV product has been driven by the industry transition to 5nm. That transition is completed, with 3nm orders delayed, per KLA: “...in the past two years it was the population of the N5 node...that was most of it...now it's N3 and N2...they're delaying those PO's.”
- Lasertec has only two potential growth drivers left - Intel and TSMC, with Samsung as a distant third. TSMC is frustrated with the tool's reliability and throughput. Its ACTIS purchases have hit the wall, not to mention the ACTIS A300 failing TSMC's evaluation. That only leaves Intel, but ACTIS tools have only two use cases: in the mask shop and in production in fabs. The mask shop needs only a few tools, with TSMC and Intel's shops at a total of a mere 5-6 systems and no additions in ages.
- That leaves only one theoretical driver of growth: Intel's potential addition of a few more ACTIS tools in its fabs, as TSMC views them as a failure for production and eliminates its fabs as a growth opportunity. We note that masks are first inspected in the mask shop, and also periodically inspected in the fab shop as wafers are printed. An EUV industry expert bluntly framed the dilemma: Lasertec's tool is simply not production-worthy, which makes two mask shops at Intel and TSMC, both now full, as the bulk of its market opportunity: “the Lasertec tool is not 95% reliable...the mask shop...that's where more of the money is...inspecting in process in the wafer fab – the importance is lower than the mask shop...that's why spending money on an EUV actinic inspection in a wafer fab seems unlikely to me...it would be lower priority overall.”

Detailed summary / key findings

8. Lasertec's small, one-time EUV market opportunity is already exhausted (cont'd)

- It is therefore no surprise that Intel and TSMC are both delaying Lasertec EUV tool orders. A semiconductor executive in Japan in contact with Lasertec's CEO stated "it has been slow...companies like TSMC were holding back...Intel was delaying their orders for more than a year." One of Intel's most senior and tenured engineers with leadership responsibility for EUV added that "TSMC is pushing out and delaying orders."
- Our research interviews indicate that Intel has purchased about 13 total ACTIS A150 EUV inspection tools. We believe 3 are installed at its mask shop in Santa Clara, CA, according to an ex-Lasertec field service engineer who was based on-site there; and we think that 10 are at its fabs in Oregon, according to an Ushio field service engineer based there. The mask shop appears to have hit capacity years ago with no new installs, leaving only Intel's fabs as a potential source of growth. However, since 2020, Intel's average rate of adding ACTIS machines at its fabs is only 2.5 per year, by our math. At a slightly discounted price of \$40MM each, Intel's fabs could drive \$100MM/year revenue at most. A Samsung executive estimated Intel would buy even less – only one machine per year, based on Intel's ratio of Lasertec tools per ASML EUV system.
- An ex-TSMC purchasing executive - still in touch with colleagues - told us that TSMC has only purchased six ACTIS EUV tools since their introduction in 2019, with no new purchases in years. At about \$50MM each – the executive complained that "the pricing is so expensive" - that is a mere \$300MM in cumulative EUV revenue in five years. We believe two of the six are in TSMC's mask shop, and the rest are in its fab where it has struggled with their pitifully low reliability and throughput. The executive was clear that its only use case is as a niche R&D tool: "it's not for production." An ex-Lasertec field engineer at TSMC's fab in Taiwan corroborated by stating that "I want to emphasize that the A150 are going into R&D at the site...mostly R&D."
- The ex-TSMC purchasing executive bluntly affirmed that TSMC is no longer purchasing the A150 EUV tool, stating "no, unfortunately no" when we asked. He repeatedly mocked it as a "lab tool, R&D tool" – "it's not a tool to support production...the Lasertec tool cannot meet demands in a production line." He stated that he recently spoke with a Lasertec executive, who admitted that things "are not well" for them.
- An Ushio field engineer who services the light sources in Lasertec's machines corroborated the color we received from the TSMC purchasing executive, confirming that TSMC bought six ACTIS EUV tools in total. He indicated that the last one was purchased in 2022. We note that the Ushio engineer is based on-site and full-time at an Intel fab in Oregon and is in touch with colleagues in Taiwan who service TSMC.

Detailed summary / key findings

8. Lasertec's small, one-time EUV market opportunity is already exhausted (cont'd)

- TSMC cannot be a growth driver or key customer going forward for Lasertec for other reasons – an alarming fact as it is the world's largest chipmaker and one of the three anchors that every inspection vendor depends on. First, Lasertec's actinic inspection system is only relevant for masks with a pellicle, and TSMC generally doesn't use pellicles – a thin membrane to protect the mask from debris. Lasertec's tool purportedly allows inspection through the membrane, although in reality it introduces debris thus defeating its purpose and requiring duplicative inspection with a KLA DUV system. Second, TSMC is not moving to ASML's new High NA system, using multi-patterning instead – eliminating High NA as a growth driver despite its hype as the next big thing in EUV.
- Given the failure of Lasertec's EUV tool in a production environment, TSMC simply relies on legacy DUV tools instead, stating that DUV is sufficient to inspect EUV masks in the fab – an ominous fact for the Lasertec EUV bull case. The ex-TSMC procurement executive stated that the only Lasertec tool they are willing to use for production is their MATRICES DUV system, not their ACTIS EUV line – "TSMC will not consider the A300 or A150 for production"; "Lasertec X8 ULTRA and X9 ULTRA are the DUV light source but they are inspecting the EUV mask...for the throughput...[EUV] is not necessary."
- Although TSMC uses a few Lasertec DUV machines in production, there is no growth story there. The procurement executive stated that "TSMC will keep a minimal share for Lasertec" only to have some purchasing leverage with KLA. He added that "two-thirds of the business migrated from originally Lasertec to KLA...I have to say Lasertec has continued to decrease...their share in TSMC."
- Legacy DUV sales will not save Lasertec from its EUV failure, as TSMC indicated that at most they may purchase 15 Lasertec DUV tools in coming years versus 10 of MATRICES X8 that they have currently. They indicated that the new X9 is still in evaluation, with an ex-Lasertec field engineer at TSMC telling us that the X9 has light source problems – thus jeopardizing even the 15 potential tool purchases.
- Samsung and Global Foundries are the only remaining buyers after Intel and TSMC where Lasertec could salvage a bit of EUV tool growth. However, executives at Samsung, Global Foundries, and a major Japanese tools supplier told us that only one Lasertec EUV tool is required for every five ASML EUV scanners installed. ASML reported a bit over 10 EUV scanners in each of the last 4 quarters: 12 (Q2/23), 11 (Q3/23), 13 (Q4/23), 11 (Q1/24) – implying at most two ACTIS EUV tools per quarter. In reality, the actual number is closer to one ACTIS tool per quarter at ~\$50MM each, as TSMC is a major portion if not a majority of ASML EUV systems – and TSMC is no longer buying Lasertec's EUV tool.

Detailed summary / key findings

8. Lasertec's small, one-time EUV market opportunity is already exhausted (cont'd)

- The Samsung executive indicated that they purchased about 10-15 ASML scanners in each of recent years, indicating a ceiling of 2-3 ACTIS EUV machines if that rate holds – which seems unlikely given the flat and now declining quarterly EUV tool shipments recently reported by ASML. Samsung stated they pay about \$40MM for each Lasertec machine, which implies \$80-120MM of revenue at most – insufficient to maintain Lasertec's growth and valuation as Samsung is one of the three legs of the customer stool they depend on.

Detailed summary / key findings

9. KLA's impending launch of an EUV actinic patterned mask inspection tool will be catastrophic for Lasertec. We spoke with 3 KLA sources, as well as senior individuals at Intel, TSMC, Samsung, Global Foundries and others – all indicated that KLA's tool is likely coming in 2025, has a superior technology, and will destroy Lasertec's market share – hardly a surprise, given previous short-lived "monopolies" in semicap equipment that collapsed overnight when leapfrogged by a new entrant. Customers irate with Lasertec are "begging" KLA and "crying" for it to hurry its launch; KLA's entry will be "devastating," "catastrophic," and "game over" for Lasertec which "will disappear" – "everything could turn around in a second...they're out of business, basically." KLA's tool uses a superior EUV source technology and is specifically timed for the transition to High NA, for which Lasertec has no capability. KLA is in pole position given Biden's CHIPS Act, which is placing massive pressure on Intel, TSMC, and Samsung to buy from KLA, given massive US subsidies just granted to chipmakers, which sources call a seismic event that spells major trouble for a Japanese supplier like Lasertec.

- KLA has been tight-lipped about its technology and plans, but recent statements by its management team underscore their seriousness. The EUV tool was hinted at several times under the name Teron "7XX," at their last investor day in 2022. At a 2023 investment conference, KLA's CFO stated they have already invested "over \$1 billion" specifically for EUV mask inspection: "If you look at the reticle space, for EUV...at the end of this year, I was adding up the numbers as CFO, you kind of roll your eyes you go holy – wow, over \$1 billion across multiple products to solve the reticle challenge in terms of R&D investment." He noted their dominance and effective monopoly in inspecting EUV masks with their Teron DUV systems: "in the EUV space, 90%-plus of reticles are running through KLA systems."
- KLA's actinic inspection tool was in development circa 2015/2017 but put on hold as they believed the market for such a tool was too small – a prescient view given the insignificant amount of Lasertec revenue that actually comes from EUV actinic tools versus its old DUV systems. When KLA scaled back its EUV mask inspection program in 2015, the CEO's comments on its earnings call were devastating regarding the lack of a market for EUV mask inspection tools: "if we build that tool and there's not a market for it..."; "we're coupled to our customers on this and talking closely with them about their needs...to date, we have not seen compelling interest on their part to want to make that investment."

Detailed summary / key findings

9. KLA's impending launch of an EUV actinic patterned mask inspection tool will be catastrophic for Lasertec (cont'd)

- KLA stated that actinic EUV mask inspection only made sense in conjunction with High NA EUV lithography – which is only beginning now, with ASML shipping the first High NA EUV scanner just a few weeks ago in April 2024 to Intel. At KLA's investor day, they stated that their actinic tool is “in development, intersecting with High NA EUV,” with an executive commenting that “we could have developed that earlier” but “we are confident that we are ready for the market at the right time...and be ready whenever high NA EUV will be available.” Given Lasertec's inspection tool lacks High NA capability, as detailed in an earlier section, KLA's patient entry into the market with High NA technology is an existential threat.
- We spoke with 3 KLA sources who indicated that its actinic tool is coming in 2025: an ex-executive in a leadership role in its EUV group, who was directly involved in its development and is in touch with colleagues; a current executive with a key role in source development; and an ex-director of engineer in the EUV group. The first executive stated the timeline was originally end of 2024, but that he believes the beta version will ship in mid-2025 with KLA accepting purchase orders in Q3 2025. He stated it's “almost a billion dollar investment”: “it's happening according to all indications I have...of course it's happening...it's not bullshit...they are investing tremendously...they have committed to Wall Street.”
- We note skepticism among some in the industry given KLA's secrecy and previous delays, which led us to press our research sources to verify that KLA's market entry is real. An ex-KLA executive stated that KLA spent years working on its actinic EUV tool and developed a prototype, and that the delays were not technological but due to a miscalculation as to when ASML's EUV scanners would launch, given that ASML had been delayed for a decade – without ASML EUV scanners, there would be no EUV masks to inspect and no market for such tools. A longtime industry consultant who is close to KLA and recently completed a consulting project for Lasertec stated KLA “made a conscious decision” to walk away from the market as no customers wanted to pay for the product.
- A second KLA executive, with extensive experience in EUV source development, confirmed their technical ability to launch the tool given their “virtual monopoly for the DUV tool”: “KLA has all the ability to make it happen...that's for sure...KLA is definitely it.” He replied “for sure” when we asked if he's confident that KLA will solve any EUV light source challenges. A leading EUV lithography expert close to KLA as well as to Lasertec customers agreed: “KLA may have an interesting strategy...they will be ahead of Lasertec because they already have a better source.”

Detailed summary / key findings

9. KLA's impending launch of an EUV actinic patterned mask inspection tool will be catastrophic for Lasertec (cont'd)

- The “better source” the EUV lithography referred to is xenon-based. The ex-KLA executive corroborated that “the KLA xenon works,” using “frozen xenon gas with IR laser as the originator,” unlike Lasertec’s Ushio or Isteq tin-based sources which he stated are a failure. He predicted that the impact on Lasertec would be devastating, with KLA winning each step of EUV mask inspection, in addition to creating brutal price pressure as customers like TSMC used KLA’s entry to crush Lasertec’s pricing: “TSMC will play hard...KLA will win the P2 and P3 [inspection]...that’s for the mask shop.” As KLA’s Teron DUV tools already have a monopoly in P1, KLA would own the mask shop.
- All three KLA sources we interviewed, as well as every major customer and other experts, stated that KLA’s market entry will be catastrophic for Lasertec’s EUV business. One stated “they will struggle...extremely hard to compete with KLA...I don’t think Lasertec can pull it out in EUV [versus KLA]” – and that “they will disappear in five years” if they don’t “re-engineer the tool [with] a simple source.” A second indicated that “people would love to buy from KLA...there is no question about that...even if [their tool] were crap, then the customer would love to have a second supplier...they will get the orders, that’s for sure.”
- One of the KLA sources noted that during his leadership role in their EUV mask inspection group, key Lasertec customers were coming “crying” to KLA and “begging us to expedite”: “I’m talking about customers coming to KLA... saying, guys, we need you to develop this tool...P2 can only be done by a clean tool...they were really begging us to expedite.” A second KLA source – an ex-director of engineering also in the EUV division – confirmed that “customers would really like KLA on this” – the Lasertec tool “has a lot of issues” and “doesn’t work well...that’s why customers want KLA to make a reliable tool.”
- Non-KLA sources reiterated the same doomsday scenario for Lasertec. A longtime semiconductor equipment executive in Japan – friendly with Lasertec’s CEO – stated that “everything could turn around in a second if KLA comes out with a superior tool...they’re out of business, basically.” The executive replied “correct” when we asked if it’s “game over” when KLA comes out with their product. He expressed skepticism of Lasertec’s stock and the average Japanese investor’s lack of understanding of EUV: “they were picked up by mass media newspapers and TV...that’s why they became popular...average people walking on the street have no idea what EUV is.”

Detailed summary / key findings

9. KLA's impending launch of an EUV actinic patterned mask inspection tool will be catastrophic for Lasertec (cont'd)

- Intel and Samsung both told us that they believe KLA is soon entering the market. One of Intel's most senior and tenured engineers, who plays a leadership role in EUV and their Lasertec relationship, provided color that ought to make Lasertec investors quake in their boots, given Intel is Lasertec's "founding" and core customer: "it's a problem to be dependent on one supplier...we always want multiple choices...the timeline you heard from KLA is reasonable...I think KLA is serious...a year or eighteen months is reasonable." A longtime ex-Samsung executive, also in a leadership role in their EUV effort until recently and who works closely with Lasertec as a decision-maker and still speaks with colleagues, echoed Intel's view: Samsung "is always trying to diversify vendors...they don't want to just rely on one company because it's not good for negotiation power...they keep pushing KLA to produce the same tool as Lasertec."
- The Samsung executive stated that KLA's tool is coming soon: "KLA kept talking like it's very soon, just in a year...I think it's pretty soon...in a year or so...KLA is working very hard to develop those...." He predicted that KLA would get half of Samsung's business, and provided a particularly interesting piece of information – indicating that Samsung has already pre-ordered the tool from KLA: "they actually made pre-orders...for evaluation and testing...Samsung basically has confidence and trust in KLA...KLA is aggressively developing their own actinic tools...they're going to find a way...one way or the other, sooner or later."
- Global Foundries view confirmed the color from Intel and Samsung, stating that "there's a huge risk here to Lasertec...if machines are piling up and KLA is going around reassuring people that if you can hold off until mid-2025." He predicted "they will take share from Lasertec...they obviously will" and that Lasertec would have to take an impairment charge on its mountain of inventory: "Lasertec could be caught with inventory that they're going to have to significantly mark down...they would end up having to impair...could put a lot of pressure on them, which would probably require an asset write-down."
- TSMC appears to have stopped purchasing Lasertec actinic EUV machines several years ago, given their lack of reliability and throughput, as detailed in prior sections. However, KLA's actinic product is still likely to smash TSMC's purchases of Lasertec's legacy DUV machines, of which it still purchases a few for inspection in the fab, despite their lower resolution vs. EUV. TSMC would likely divert those purchases to KLA's actinic product versus trying to improvise with Lasertec's DUV system. An ex-TSMC procurement executive bluntly stated they already have a much higher opinion of KLA than Lasertec: "for productivity and for economic considerations, KLA performs better than Lasertec...two-thirds of the business migrated from originally Lasertec to KLA...Lasertec has continued to decrease...their share in TSMC...KLA's share continues to increase."

Detailed summary / key findings

9. KLA's impending launch of an EUV actinic patterned mask inspection tool will be catastrophic for Lasertec (cont'd)

- The TSMC executive continued that, unlike Lasertec, “KLA improved a lot...they have a very good scan, and now their resolution is getting much, much better.” A second TSMC source, who was previously head of R&D and played a key role in their 5nm transition and vendor/tool selection, stated that the current transition to High NA EUV mas inspection spells more bad news for Lasertec, as Intel is under a lot of pressure to use KLA instead. We note that Intel has made the biggest bet of any chipmaker on ASML's new High NA scanners, recently taking delivery of the first High NA shipment: “Are they going to use Lasertec? I have some doubts because they were under a lot of pressure to use KLA...if they go to KLA...there will be more bad news for Japan.”
- TSMC's ex-head of R&D stated that President Biden's CHIPS Act is fast becoming a major threat to Lasertec, stating that it will force recipients of US government subsidies to purchase tools from US suppliers – the second that KLA launches its actinic EUV tool and creates a competing option, Lasertec as a Japanese supplier is in major trouble. The CHIPS Act became law in 2022 and authorizes ~\$300 billion in funding and subsidies for chip manufacturing on US soil. Intel received \$8.5 billion in March 2024, the largest grant to date – illustrating the pressure it will be under to buy tools from US players like KLA; TSMC received \$7 billion the same month, to fund a fab in Arizona; and Samsung received \$6.4B. We note that these three chipmakers comprise almost all of Lasertec's revenue.
- TSMC's ex-head of R&D described the US CHIPS Act as a seismic event – “this is actually super, super big time for suppliers”; “Japanese suppliers are in a bit of trouble...KLA definitely has a super advantage...the Lasertec guys are going to be put into second choice...KLA is very, very strong...in terms of EUV...under the CHIPS Act...KLA becomes the favorite.” He continued that “Intel, Samsung, TSMC are under pressure from the US government to use US suppliers...Lasertec, I bet they will suffer....they may lose some of these things in the US because of the CHIPS Act.”

Detailed summary / key findings

10. Lasertec's actinic EUV tool is so defective and unreliable that TSMC, Intel, and Samsung have all quietly substituted non-EUV alternatives for inspecting EUV masks, improvising with KLA's legacy Teron DUV tool plus electron beam inspection - indicating that actinic EUV is useful if and when it works but is not essential and certainly not a "monopoly" that justifies Lasertec's absurd valuation. Sources indicate that customers – presumably Intel - who claim to use its EUV tool are secretly using its old DUV system instead; that TSMC is running parallel, redundant inspections with non-EUV tools and has reverted to the DUV-based inspection process it used prior to Lasertec; that the market window for actinic EUV mask inspection is rapidly closing, as the next phase in EUV is High NA, which requires an electron beam solution, as OPC patterns and sub-resolution assist features (SRAF) are smaller than 13.5nm and cannot be detected with actinic EUV light – leading ASML, KLA, and others to develop multi-beam inspection tools for High NA masks that render the ACTIS tool obsolete.

- Industry sources told us that all three of Lasertec's key customers have quietly built their own mask inspection systems as a workaround, using a very low power EUV source called High Harmonic Generation (HHG). One source close to these customers stated his surprise at learning that they were actually using Lasertec's old DUV tool while claiming they were using their EUV machine – we suspect Intel, whose EUV mask team we think is engaging in a face-saving cover-up, given their massive bet on Lasertec. A Japanese semiconductor executive close to both TSMC and Lasertec stated that TSMC is running parallel, redundant inspections with non-EUV tools: "...parallel inspections of the same masks...just to make sure there are no defects."
- The executive stated that TSMC has reverted to the EUV mask inspection process it used before Lasertec launched its actinic EUV tool in 2019/2020: 1) using KLA's Teron DUV tool with a 193nm light source; 2) recording the coordinates of potential defects it identified on an EUV mask; 3) transferring the coordinates to an electron beam inspection tool, also known as SEM for scanning electron microscope. DUV can't distinguish very small defects on EUV masks, so e-beam is used to then analyze "these suspicious-looking defect locations...so they need a combination of optical DUV and electron beam to inspect the mask."
- The expert added that he is friendly with the CEO of Holon - a Japanese company that makes the e-beam tool used by TSMC and that it is rapidly gaining adoption among all EUV players including Intel and Samsung, indicating all three of the customers that comprise most of Lasertec's revenue have soured on its actinic EUV tool: "TSMC was the first one to adopt the Holon tool...Samsung, Intel, and most of the other customers in the EUV space are also adopting the Holon tool." The volume of Holon tools being ordered by Lasertec's EUV customers illustrate the failure of its flagship product: "I'm not quite sure... they have succeeded in phasing in Lasertec tools...they are still ordering considerable quantities of Holon inspection machines."

Detailed summary / key findings

10. Lasertec's actinic EUV tool is so defective and unreliable that TSMC, Intel, and Samsung have all quietly substituted non-EUV alternatives for inspecting EUV masks (cont'd)

- Holon's technical literature actively promotes its scanning electron microscope tool for EUV mask inspection – "becoming indispensable for EUV lithography." The semiconductor executive stated that Holon's tool only costs \$5 to \$7 million, versus \$40-60MM for Lasertec's actinic EUV tool. While KLA's DUV Teron tool is also expensive, its mask inspection throughput is vastly higher thus requiring a much smaller number of machines – TSMC complained to us that Lasertec's system could only inspect a pitiful 5 EUV masks per day, out of the 550 masks their fab needed inspected daily. Thus, the combination of KLA's DUV tool plus Holon's \$5-7MM SEM tool is far less expensive. TSMC would theoretically require 110 of the Lasertec tools at \$50MM each to cover their daily mask throughput of 550 – implying an absurd \$5.5 billion of spend.
- An ex-KLA director of engineering in its EUV mask group – who worked on each of their DUV, actinic EUV, and e-beam projects - stated that non-EUV methods like KLA's Teron DUV tools are perfectly sufficient as "alternatives for the inspection." While actinic mask inspection in theory "is the best way to do that" – using EUV so that the wavelength used in mask inspection is the same as that used to write the mask – it's only true "if the system is well-engineered," which the Lasertec system is not. He pointed to "an award announced at last year's EUV mask event about the work that KLA and others have done about supporting EUV mask inspection with other methods" – "a well-engineered system is more important than the wavelength"; "the DUV tools actually do work [for EUV masks]...those tools work really well and have been well-engineered for a long period of time."
- The ex-KLA engineering director stated that if the Lasertec tool is "not working...people can get by" using DUV, exposing that Lasertec's so-called "monopoly" in actinic EUV tools is irrelevant as DUV systems are sufficient – "KLA tool for DUV is the market leader...the mainstream tool...even for some of the EUV." We note that KLA's website actively showcases its Teron DUV systems for EUV mask inspection.
- A leading EUV lithography expert who interacts closely with all key players - Lasertec, Intel, KLA, and others - told us that all three of Lasertec's key customers have quietly built their own EUV mask inspection systems as a workaround given its defective tool, using a low power EUV source called High Harmonic Generation (HHG) - they "made essentially their own...in-house inspection system...so did Intel...Intel doesn't talk about it, but they did"; "they all ended up essentially using the same kind of technology...high harmonic generation...the power is very, very small...everybody was doing this, some quietly and some more publicly...Samsung still uses it...so does TSMC...most likely Intel does it too...but Intel is very quiet about what they do."

Detailed summary / key findings

10. Lasertec's actinic EUV tool is so defective and unreliable that TSMC, Intel, and Samsung have all quietly substituted non-EUV alternatives for inspecting EUV masks (cont'd)

- TSMC, Intel, and Samsung's use of DUV, electron beam, and HHG systems indicates that Lasertec's market window for actinic EUV mask inspection is rapidly closing, as the next phase in EUV is High NA lithography, with Intel taking delivery of the first ASML High NA scanner last month. As detailed in an earlier section, Lasertec's technology cannot support High NA – and its claims around the purported High NA capabilities of the “new” ACTIS A300 lack credibility. High NA mask inspection requires an electron beam solution similar to Holon's. The Japanese semiconductor executive friendly with Lasertec's CEO explained High NA's requirement for optical proximity correct (OPC): “actinic is not going to be sufficient for mask inspection because sub-resolution assist features will become necessary...those features you cannot detect with actinic light...ultimately an electron beam solution is required...they need features on a mask that are smaller than 13.5nm...higher resolution than EUV is required...and people are working on an electron beam solution...KLA is one of them.”
- We note his comments that KLA is working on an electron beam solution, consistent with KLA's comments that it is timing its actinic EUV product to “intersect” with High NA. KLA appears ready to leapfrog and make Lasertec obsolete as the transition to High NA gains momentum in 2024 – Lasertec cannot support High NA, while KLA has patiently prepared for it as the next phase of EUV lithography. We find this particularly ominous as Intel is the first and most aggressive adopter of ASML's High NA technology – and Intel is Lasertec's “founding customer” and was its only hope for future actinic EUV purchases. The Japanese semiconductor executive stated that Lasertec's window is “maybe another two years”: “KLA is working on it...Nuflare is working on a tool and ASML has a division where they're working on a multi-beam inspection tool...which means it has much better throughput compared to the Holon inspection tool.”
- A leading EUV lithography expert who interacts closely with all key players - Lasertec, Intel, KLA, and others - told us that all three of Lasertec's key customers have quietly built their own EUV mask inspection systems as a workaround given its defective tool, using a low power EUV source called High Harmonic Generation (HHG) - they “made essentially their own...in-house inspection system...so did Intel...Intel doesn't talk about it, but they did”; “they all ended up essentially using the same kind of technology...high harmonic generation...the power is very, very small...everybody was doing this, some quietly and some more publicly...Samsung still uses it...so does TSMC...most likely Intel does it too...but Intel is very quiet about what they do.”

Detailed summary / key findings

11. Lasertec's new "Innovation Park" in Yokohama is a fraud – a purported R&D and production campus with two supposed "fabs" that is 5X the size of its headquarters and its next driver of growth, announced in 2022 as the largest capital expenditure and expansion in its history. Multiple investigators visited the campus ~20 times from Dec 2023 and May 2024 and found a generally deserted and empty facility with no evidence of R&D, production, nor any indication that Fab 4A matches its promotional photos, leading us to wonder if the pics are fake. We conclude that the company's statements - in press releases, earnings calls, presentations, and an in-person IR meeting - are false and an epic scam, based on extensive surveillance including field visits/interviews; 24/7 time-lapse and motion-detector camera footage over several weeks; and aerial drone surveillance on two occasions. We also hired a local IR consultant who met with the company in March – Lasertec IR agreed to a brief tour of their clean room at headquarters, but strongly refused multiple requests for a tour of the clean room (Fab 4A) at the Park. We then asked for a brief tour of the Park without entering Fab 4A – also refused. Finally, we asked for simply a video clip of Fab 4A, which was also denied.

- Across ~20 field visits, our investigators found that the 5-building complex was generally deserted and empty, except for one floor of a small 4-story office building that appeared to house low-level clerical or back-office staff, not R&D. Our investigators never saw any security or guards during any visit – not what one would expect from a facility with cutting-edge R&D, sensitive industrial secrets, intellectual property, and the production or components of \$50MM pieces of capital equipment. Our investigators saw no signs or gates forbidding entry onto the complex; wandered freely; were never asked who they were or asked to leave; and in fact were cheerfully welcomed by the occasional maintenance personnel, who were happy to answer questions about the status of the site.
- We were unable to be invited inside "Fab 4A," and we conclude – as we show shortly via extensive photographs – that, at best, it is an overflow storage facility for the mountain of finished goods inventory overflowing at headquarters; but that no R&D, production, assembly, or testing takes place at the facility, even if the photos in Lasertec's presentations accurately depict the interior. In addition to Fab 4A – which the company refers to as the "first round of construction completed in June 2023," we saw no evidence of meaningful construction for Fab 4B, which it calls "the second round of construction" that supposedly began in Oct 2023 – just an office space with a ladder and few pieces of wood on the floor.

Detailed summary / key findings

11. Lasertec's new "Innovation Park" in Yokohama is a fraud (cont'd)

- We hired a local investor relations/corporate access consultant who met with Lasertec's Investor Relations and Corporate Planning department at their headquarters on March 13, 2024. Prior to the meeting, Lasertec's IR representative agreed to a brief tour of their clean room at headquarters, but strongly refused multiple attempts asking for a tour of the clean room (Fab 4A) at the Innovation Park – we find it unusual that one clean room was offered but the other refused. We then asked for a brief, general tour of Innovation Park without entering Fab 4A – also refused. In a final attempt, we asked for simply a video clip of Fab 4A, which was also denied. We were told that a shareholder flew from Switzerland to see Fab 4A and was "extremely unhappy" at being similarly denied even a video. We quote from our consultant's write-up of the meeting:
- Our request to see the finished goods bay at headquarters was also refused – given color we received from customers that it was so full with inventory that it had no space left. However, in a telling comment, the IR rep confirmed its role as a storage facility for excess finished good inventory.
- We note that the 2022 capital expenditure for the Innovation Park was by far the largest in Lasertec's history, dwarfing any other spending since 2000 – illustrating its level of importance. We show the annual spending in both JPY and USD.
- The timing of the announcement was suspicious. The stock had collapsed rapidly over 6 weeks, falling 40% from a peak of 36,000 in Jan 2022, to 21,000 on the day of the release on Feb 25, 2022 – a highly unusual moment to announce an expansion 5 times the size of headquarters, especially after sales plummeted that quarter. We suspect that management panicked and concocted a press release in a bid to halt the decline. Fake capital expenditure press releases are a trick we see often in corporate frauds, to create the illusion of customer demand, expansion, and growth.
- We note three other unusual circumstances around the Park, in addition to its suspicious timing. First, the press release at closing stated that Toshiba used the site and its 5 buildings as a training center – but that Toshiba would continue to use the facilities for another year after the closing of the purchase. We have never seen a company make the largest capital expenditure in its history, and then agree to not use it for a year – especially if it was meant to "respond to customer demand."

Detailed summary / key findings

11. Lasertec's new "Innovation Park" in Yokohama is a fraud (cont'd)

- The second troubling circumstance: although it was a Toshiba training facility, it was owned by Nomura Real Estate Holdings, which had purchased the facility from Toshiba in two tranches – and Lasertec explicitly declined to disclose it as a related-party transaction, stating "Nothing of note" for "Existing relationships between the seller and Lasertec." Yet Nomura Asset Management is the second largest holder of Lasertec with 7%. We note only that both Nomura and Toshiba have an extensive history of episodes involving serious fraud. Nomura's real estate group, in particular, seems to be a compulsive repeat offender. In 2018, the US Department of Justice announced a \$480MM settlement with Nomura for "fraudulent conduct in connection with its Residential Mortgage-Backed Securities offerings." In August 2023 – less than a year ago – the DOJ announced a new \$35MM settlement with Nomura and the resolution of potential criminal charges after it "perpetrated a scheme...to defraud its customers...with the knowledge, encouragement, and participation of supervisors" – once again with respect to mortgage-backed securities.
- Third, we were unable to locate any construction or other permits for Lasertec's purported build-out activity at the Innovation Park. Our investigators visited the City of Yokohama's Housing And Architecture Bureau, and the only permits we could find were from 1997 for Toshiba's construction of the four buildings, other than Building A, which dates from 1969.
- We document a long pattern of specific claims by Lasertec management regarding InnoPark that are false: "a clean room has been newly established...and is being used as a research and development base from July 2023"; "the new InnoPa development site has been in use since July for product assembly and final adjustments" and that they would "further increase production volume" there "once the second phase of construction is completed"; "we have not announced...how much production will expand"; "some clean rooms have been installed"; "five times the space of our current headquarters"; that "production began in July"; that "equipment is now being assembled"; "will contribute to production...we are working to increase production capacity...expanding our production space and workforce"; "working to increase production capacity."
- We monitored the entrance to Fab 4A with a 24/7 time lapse camera, for several weeks from several angles – we show photos from different days and times. We observed almost no activity, except for an occasional machine being delivered, as it serves only as a storage facility for finished goods inventory, establishing that it Inova is not staffed and is empty of people. We saw no staff entering or exiting at typical peak times such as morning, lunch, or after work. We saw only an occasional maintenance worker or vehicle, or delivery of a finished machine.

Detailed summary / key findings

11. Lasertec's new "Innovation Park" in Yokohama is a fraud (cont'd)

- Our investigators were allowed to wander freely around the complex without anyone asking them to leave. During one visit, they stood in front of Fab 4A as machines were being delivered. At that moment, we came to understand the only purpose that the so-called fab serves: as a storage facility for overflowing finished goods inventory. Photos show that the machines are already shrink-wrapped and hermetically sealed in extensive protective wrapping, and that they are thus finished good with absolutely no testing or fine-tuning inside Fab 4A, as claimed by management. Furthermore, the machines below appear far too small to be Lasertec's DUV or EUV systems – we assume they are their old, low-cost tools.
- We contracted an investigator for aerial drone surveillance, who took videos and photographs through the windows located at the top of Fab 4A on two separate days. We were curious if any clean room exists at all inside Fab 4A, similar to the photo of the large, new, bright-white, extensively-lit room shown in Lasertec's presentations. From the top of Fab 4A, the drone could see straight through to sunlit windows on the other side of the structure, which still looks exactly like a gymnasium. The interior looks like a large open space with an exposed scaffold ceiling for HVAC equipment, as one would expect for a large gym-type space. We saw no evidence of a large clean room like the one in Lasertec's photo.
- Clean rooms for semiconductors and inspection equipment are the cleanest facilities on earth, where one microscopic particle discovered in a large facility every few weeks is considered a problem. Yet, the windows for Fab 4A are filthy and covered in dust. The HVAC equipment visible behind the windows appears caked in rust and/or debris. Close ups of mechanical and/or HVAC structures inside Fab 4A look like those in a ghost town gathering dust and spider webs for years.
- In addition to Fab 4A, where construction was supposedly completed June 2023 when it became "operational," Lasertec states that the "second round of construction" began in Oct 2023 for Fab 4B – but investigator site visits indicate that these claims are also a wild exaggeration and fraudulent. However, the entrance to Fab 4B indicates it is just an office space. Our investigators were able to easily look through the windows to "Fab 4B" to see the level of construction activity. They saw nothing but a ladder and a few pieces of wood on the floor, and no construction crews. No signs of progress for a fab are visible – not what one would expect if a major phase of construction began 8 months ago.